# Benchmarking MICA against ModelAngelo, EModelX(+AF) and newer deep-learning model builders on cryo-EM maps released in 2026

Muhammad Zohaib
BS Bioinformatics, Department of Computer Science, University of Agriculture Faisalabad, Pakistan
zohaib.bioinfo@gmail.com · GitHub: Zohaib-Bioinfo
Practical task for the research-assistant selection of Prof. Jianlin Cheng · Report date: 30 October 2026
Code, entry list, environments and raw per-entry metrics:
https://github.com/Zohaib-Bioinfo/mica-benchmark

**Abstract**

MICA builds protein models from cryo-EM maps by combining the density with AlphaFold 3 (AF3) predictions. I benchmarked it on 20 single-particle entries whose maps and models were released in 2026 (resolution 1.24–3.50 Å, 173–682 modelled residues) against ModelAngelo, EModelX(+AF) and two newer deep-learning methods, EMProt and E3-CryoFold. The entries were chosen by rules fixed before any run; every method ran with the settings of its own README on free Kaggle machines, the methods with an AF3 input got the same predictions, and one script scored every model against the deposited model. MICA built a model for 17 of the 20 entries (two runs were killed for lack of RAM, one reached its time limit); its median TM-score was 0.94 (mean 0.94, 95 % CI 0.91–0.96). On the entries where both built a model (differences are method minus MICA), EModelX(+AF) did not differ from MICA (0.00, 95 % CI −0.02 to 0.03), EMProt was slightly higher (0.04, 0.02 to 0.06; Holm $p < 0.001$), E3-CryoFold was lower (−0.08; Holm $p = 0.020$) and ModelAngelo was lower on the maps above 3.0 Å (−0.30, −0.48 to −0.13; Holm $p = 0.059$) but not on the maps up to 3.0 Å. Free compute forced two deviations from MICA's own pipeline, a coarse docking search and the omission of its real-space refinement, which the Phenix version available here would not run; both are described with their probable effects. The sample is small, and the conclusions apply to maps of up to 682 modelled residues. Code, the frozen entry list, environments and the raw per-entry metrics are public.

## 1 Introduction

Building an atomic model of a protein from a cryo-EM density map is still largely manual. MICA (Gyawali et al., 2025) automates it by combining the map with structures predicted by AlphaFold 3 (AF3; Abramson et al., 2024), and its authors report a clear advantage over ModelAngelo (Jamali et al., 2024) and EModelX(+AF) (Chen et al., 2024), including a mean TM-score of 0.93 on a set of recently released maps. This report asks a narrower question that a reader can check: *how well does MICA build protein models on maps that were released in 2026, and how does it compare with the same two baselines and with newer methods, when everything runs on free compute and every method gets the same inputs?*

The test set is made of EMDB maps and PDB models that were both released between 1 January and 12 October 2026 (the freeze date). Twenty entries were chosen by a rule that was fixed before any model was run (Section 3). Five methods were run on all twenty entries with default settings and no per-map tuning (Section 2); every output was scored in one session with one set of tools against the deposited model (Sections 4 and 5). Failed runs stay in every table, and a sensitivity analysis scores them as zero. With about twenty entries the report puts effect sizes and confidence intervals before $p$-values.

**Findings in brief.** MICA built a model for 17 of the 20 entries (median TM-score 0.94). The other methods were equal, slightly higher or lower depending on the method and the resolution;

**Table 1.** The methods and how they were run. All commits are pinned in `methods/methods_table.csv` and in the run information of each session. The scored file is the one fixed before any score was seen (Section 5).

| Method | Paper | Commit | Inputs used here; file that is scored |
|---|---|---|---|
| MICA | Gyawali et al. (2025) | `726d1bb14b` | Map and one AF3 prediction per unique chain, docked into the map by MICA's program 2 (Section 2.2). Scored: `MICA_all_atom_model.pdb`. |
| ModelAngelo | Jamali et al. (2024) | `994945bdfa` | Map and sequence; no AlphaFold input (`model_angelo build`). Scored: the sequence-assigned model. |
| EModelX(+AF) | Chen et al. (2024) | `53713cf820` | Map, sequence and the AF3 predictions as templates (protocol `temp_flex`, no AlphaFold database download). Scored: `all_atom_model.pdb` (after Pulchra). |
| EMProt | Li et al. (2026) | `9489d52e7f` | Map, sequence and the AF3 predictions as per-chain models (`emprot build ... -chain`). Scored: `output.cif`. |
| E3-CryoFold | Wang et al. (2025) | `241cef48da` | Map and sequence (`inference.py -protocol pre_align -t 0.1`); no AlphaFold input. Scored: the `pre_align` model. |

only the difference to EMProt is clear on both statistical tests (Section 6.3). Two deviations from MICA's own pipeline, forced by free compute, and the small sample limit what can be concluded (Section 7).

## 2 Methods

### 2.1 The five methods

MICA is the method under test. Two baselines are the ones that its paper compares with, ModelAngelo and EModelX(+AF); two are newer deep-learning model builders, EMProt and E3-CryoFold, that passed a published screen (Section 2.4). Every method ran with the settings of the example in its own README, with no per-map tuning. Table 1 gives the version, the inputs and the file that was scored for each. All methods that accept predicted structures got the same AlphaFold 3 (AF3) predictions; ModelAngelo and E3-CryoFold, which have no such input, got the map and the sequence only.

### 2.2 MICA and how it was run here

MICA combines the cryo-EM map with AF3 predictions in three programs (Gyawali et al., 2025). Program 1 splits every AF3 prediction into domains (Merizo; Lau et al., 2023). Program 2 docks each domain into the map with Phenix `dock_in_map` (Liebschner et al., 2019). Program 3 resamples the map to 1 Å voxels, predicts maps window by window with a 3D neural network (windows of $64^3$ voxels, step 48), which takes the docked AF3 models as extra input channels, builds the chains from these predictions and from the AF3 structures, completes the side chains with Pulchra (Rotkiewicz and Skolnick, 2008) and refines the model against the map with Phenix real-space refinement (Afonine et al., 2018). MICA's code and weights are used as released (commit `726d1bb14b`); the network, the building of the chains and the refinement are not changed. Four things differ from the way the authors run MICA, each forced by free compute; the evidence is below or in Section 4.4, and the effects are discussed in Sections 6.6 and 7:

1. **Docking.** On a test entry (26DK, one chain of 191 residues, 3.5 Å) MICA's own docking call (Phenix molecular replacement, no time limit) did not finish in 66 minutes on a 4-core, 31 GB Kaggle machine and was then killed by the system for lack of RAM. Program 2 is therefore run through a small driver that imports MICA's own file and replaces only the method that calls Phenix: `dock_in_map` in its Fourier-search mode with `quick=True`, at 6 Å (then 10 Å), with a time limit per domain. The placements are coarse (Section 6.6).

2. **No docked input where nothing can be docked.** If no domain of an entry docks, MICA still runs without docked AF3 input, which its README describes as an option ("skip this step if you don't want to use AF3 at input level"); the AF3 channels of the network are then zero. The AF3 structures are still used when MICA builds the chains. Every MICA result is labelled by its docking mode: *full* (every domain docked), *partial* or *none*.
3. **Program 3 on a GPU, one window at a time.** MICA predicts windows in batches of up to 8 when a map has more than 200 windows. That setting (`batch_threshold = 200` in `utils/predict.py`) exceeded the 15 GB of a T4 on the first three large maps. In the GPU sessions of the second and third round it is raised to $10^6$, so that every window is predicted alone, as MICA does for small maps. This changes memory use only.
4. **No real-space refinement.** MICA's last step, Phenix real-space refinement with the parameter file shipped in MICA's repository (written for Phenix 1.20.1), stops with a parameter error in the only Phenix version available here (2.2.1-6174: "Incompatible parameter objects: scope `cdl_nucleotides`"). It stopped with this error in every one of the 25 program-3 runs of the GPU sessions that built a model (the main sessions and the session without docked input; the run logs are in the repository), while the program itself reports exit code 0. The scored file is therefore MICA's all-atom model after Pulchra and before refinement, chosen before any MICA result existed because it is the program's own output. Refinement moves atoms only a little and is not expected to change the TM-score much, but its effect on the other metrics is not measured (Section 7).

### 2.3 The baselines

*ModelAngelo* (Jamali et al., 2024) was run as `model_angelo build -v MAP -pf FASTA -o OUT -device 0` with the default weights bundle. *EModelX(+AF)* (Chen et al., 2024) was run in its AlphaFold mode (`-protocol=temp_flex`) with the AF3 predictions as the templates of its README example (one folder per sequence with `ranked_0.pdb`), followed by Pulchra. Feeding AF3 models to it is a protocol choice made so that all methods with an AlphaFold input get the same predictions. *EMProt* (Li et al., 2026) was run as `emprot build -map MAP -seq FASTA -chain <AF3 models> -device 0`. *E3-CryoFold* (Wang et al., 2025) was run with the `pre_align` protocol of its README example; its `denovo` protocol is not part of the benchmark.

### 2.4 Selection of the newer methods

The newer methods were chosen with a screen that was written down before the search (spec decision 2): a method had to be published after both baselines, use deep learning, build protein atomic models from a map and a sequence, have public code and weights, run on a free 16 GB GPU and have documented defaults. A literature and repository search (to 11 October 2026) screened 29 methods: 3 passed (CryoAtom, EMProt and E3-CryoFold), 2 were kept in reserve (EM3DFold, a preprint; CryoZeta, which needs 32 GB of GPU memory) and 24 were excluded, each with a reason (Appendix, Table A1). CryoAtom (Su et al., 2026) was dropped before the production runs, as the first scope cut of the spec allows: in a pilot on the smallest entry neither of its two versions finished within 2 hours, and at the floor of 12 entries it would have needed more GPU hours than one weekly quota. The benchmark therefore has two newer methods, not three. The decision did not depend on any result. Forward citations of MICA could not be searched (the citation databases were not reachable from the workspace), so the screen may have missed methods that only a citation search finds.

## 3 Datasets

### 3.1 Selection of the entries

The test set was fixed before any method was run (spec decisions 3 to 5; the entry list was committed on 16 October 2026). An entry qualifies when its EMDB map and its PDB model were both released between 1 January and 12 October 2026, so that no method can have been tuned on it by the

authors of the maps. The release date of every map was checked in the EMDB, and the pool was frozen on 12 October. The filters were applied to the metadata and the files of the deposition:

- single-particle reconstruction, resolution of 4.0 Å or better, one EMDB map per entry;
- protein only (no nucleic-acid chain), 1 to 700 modelled residues in total, at most 3 unique chain sequences, at most 5 % UNK residues;
- a map–model correlation $CC_{mask}$ of at least 0.70, computed for every entry in the same way with Phenix `map_model_cc` at the resolution of the entry.

Table A2 (Appendix) shows how the pool of 6 116 cryo-EM entries released in the window shrinks to 484 candidates. The candidates were clustered with MMseqs2 (Steinegger and Söding, 2017) at 30 % sequence identity on the longest chain of each entry (179 clusters), so that no protein family is used twice. Two strata were defined by resolution (at most 3.0 Å, and above 3.0 up to 4.0 Å). In each stratum the candidates were put in a random order (seed 2026) and the first 10 that passed every filter were taken; an entry whose cluster or map was already used was skipped. The $CC_{mask}$ rule removed 6 entries (all in the lower-resolution stratum) and 2 more because Phenix could not compute it (a charged-oxygen atom type that its scattering table does not know). The seed, the shuffle and every skipped entry are recorded in the selection log of the repository.

### 3.2 The twenty entries

The final list has 20 entries, 10 per stratum (Table A3 in the Appendix). Resolutions range from 1.24 to 3.50 Å; the deposited models hold 173 to 682 residues (median 454); 13 entries have one protein chain and 7 have several chains, some of them copies. The maps range from 16 MB to 1 GB. Because the rule takes one entry per cluster and the seeded order decides which one, the list favours common specimens only to the extent that the pool does (seven clusters hold a quarter of the candidates); the cluster sizes of the chosen entries are in the repository.

**How new is the protein?** The maps are new, but the proteins often are not. For every unique chain we recorded the highest sequence identity to any chain of a PDB entry released before 2026 (MMseqs2 search against `pdb_seqres`, alignments covering at least half of the chain). 13 entries have a chain with 95 % identity or more to an older entry (for example 22FX, human apoferritin), 4 lie between 44 and 76 %, and only 3 (9TKA, 9VBD and 26DK) are at 0 %. Training data of the methods are not known in detail, so a method may have seen a homologue of a test protein. The planned sensitivity analysis on the subset at 30 % identity or less has 3 entries, too few for a test; it is described in Section 6 and not tested.

### 3.3 AlphaFold 3 inputs

One AF3 prediction was made for each of the 26 unique chain sequences of the 20 entries with AlphaFold Server (default settings, random seed, no custom templates); the first model (`model_0`) is used by every method that takes predicted structures. The mean pLDDT of the predictions ranges from 62.4 to 97.5 (median 87.0); the chains with most residues below 50 are 9VUH chain E3 (43 %), 9V4D chain E1 (25 %) and 9VUH chain E2 (22 %). A nine-residue peptide with an unknown first residue (9V4D chain E2) was submitted with glycine in its place. Each prediction was checked against its submitted sequence. The predictions are not published in the repository (the Server terms allow non-commercial research use only); the job list, the checks and the file hashes are.

## 4 Benchmarking procedure

### 4.1 Compute and software

The pool of 2026 releases and the map–model correlation of the candidates (Section 3) were computed in free Google Colab notebooks; every run of a method and all scoring ran on free Kaggle notebooks. No paid service was used. A GPU session has two NVIDIA Tesla T4 GPUs (15360 MiB each), 4 CPU cores and 31.3 GB of RAM; a CPU session has the same cores and memory

and no GPU. Phenix 2.2.1-6174 was installed with the author's academic licence (the installer is not redistributed); US-align and every method were installed from pinned commits (Table A4). Each session wrote the output of `nvidia-smi`, the number of cores, the memory, the package versions and the hashes of its input files into its result zip. The notebooks are generated by scripts of the repository and were tested against stand-ins of the methods before they were run; the sessions that produced results are listed in Table A5. A result zip is unpacked by a script that keeps text files only (coordinate files stay in the notebook outputs).

### 4.2 Order of work

(1) The entries were selected and frozen, and the AF3 predictions were made (Section 3). (2) A pilot of every method on the smallest entry (26DK) fixed versions and commands before any production run. (3) The four non-MICA methods ran on all 20 entries in three batch sessions. (4) MICA ran in three stages: Merizo domain splitting and docking in CPU sessions, then the network and the model building in GPU sessions. (5) All models were scored in separate CPU sessions with one notebook code, using the protocol of Section 5, which was written down on 24 October, before any MICA model existed. No entry was dropped or replaced after its first run, and no parameter was changed for a single map.

### 4.3 Rules for the runs

The rules below were written in the project log before the sessions that used them started; Table A8 lists every failure with its reason.

- **One run per method and entry.** A run that ends without the model file counts as failed, whatever the reason (crash, out-of-memory kill, timeout). Failed runs stay in every table and are scored as zero in a sensitivity analysis (Section 6).
- **Run limit.** 3 h for every run of the baselines; for MICA, 30 min for program 1, a limit for each docking call (Section 4.4) and 2 h (GPU) or 3 h (CPU) for program 3. Reaching a limit is a failure. The spec allowed 11 h per run; it was shortened because a weekly GPU quota of about 30 h could not carry hung runs.
- **Two programs at a time.** A Kaggle GPU session has two GPUs; the baselines ran one program on each. Their run times are therefore wall-clock times with another program active, and only indicative. E3-CryoFold and EModelX(+AF) never ran at the same time, because both allocate arrays that grow with the square of the number of candidate atoms and ran out of RAM together in the pilot. A baseline that the system killed (exit code $-9$) while the other GPU was busy was run once more, alone; a kill while alone, a crash or a timeout is a failure at once.
- **Cut by the end of a session.** A run that the end of a session interrupted is not a result; it was run again in a later session.
- **Order.** The baselines took the entries in the seeded order with the two strata interleaved, so that any prefix is balanced; nothing was dropped or left out because of a result.

### 4.4 MICA in three stages

**Docking (program 2).** The AF3 prediction of every chain was split into domains by Merizo (program 1; 59 domains in the 20 entries). Docking was run three times, always on CPU sessions of at most 12 h, and the three passes are all reported (Table A6). *Pass 1* called Phenix `dock_in_map` with `dock_with_mr=False quick=True` and at most 45 min per domain, followed (if the quick search had not run out of time) by a thorough search of at most 60 min; it docked 14 of 59 domains, with a median local correlation of 0.44. *Pass 2* was meant to search at coarser resolution, but Phenix 2.2.1-6174 rejected the parameter that had been added (a True/False parameter where a number was passed), so the first two attempts stopped after one second and only the call of pass 1 ran again: 12 domains. This was an error in the set-up of the pass, found in the logs and recorded in `LOG.md`; it is the reason for a third pass. *Pass 3*, the set-up that counts, runs the whole docking, the search and the rigid-body refinement, at 6 Å (parameter `resolution=6.0`, at most 40 min per domain) and at 10 Å if that fails or runs out of time (25 min). It docked 46 of 59 domains, with a

median local correlation of 0.09. That value is not comparable with the 0.44 of pass 1, because it is measured at 6 Å after a coarse refinement; I did not check the placements against the deposited models, and the only evidence that they help is indirect (Section 6.6). By entry, 14 entries have every domain docked (*full*), 5 have some (*partial*) and 1 has none (*none*; 9VI4).

**Program 3 on a GPU.** Program 3 of MICA deletes and rewrites its own output folders, so the notebook runs it for one entry at a time. A guard switches program 3 off for the rest of a session after two failures in a row, and the entries that wait then are marked *deferred* and run in a later session. The first GPU session used MICA's own batching and failed with a GPU out-of-memory error on the maps with many windows (Section 2.2); from then on `batch_threshold` was raised to $10^6$ at run time, which was decided and logged before the rerun, and the notebook stops if the patch does not apply exactly once. In the first session 3 entries (22FX, 9WQ6 and 9V4D) failed; they ran again with the patch. This is an exception to the rule of one run per entry, which the spec allows only for runs killed by the platform. The first failure stays in Table A7, and the main analysis uses the rerun; if the reruns counted as failures, MICA would have 14 models (Section 6). Three entries have no MICA model after all sessions: 26DK and 9VI4 were killed by the system when the model-building step of MICA, which runs on one CPU core, used up the 31 GB of RAM (about 33 000 and 32 000 candidate atoms, as counted in the program logs), and 9OC0 reached the 2-hour limit of program 3 while it was filling the last gap of its last chain. Nothing else was running when 26DK and 9VI4 were killed.

**A second MICA run without docked input.** To get an indication of what the docked input is worth, a second GPU session ran program 3 from the same saved state with the docked AF3 models removed, so that the AF3 channels of the network are zero. It covers the 8 entries that the first GPU session built (Section 6.6); the equivalent sessions for the later entries were not run, because the GPU quota did not allow it.

### 4.5 Scoring and analysis

Every model file that a method wrote was scored in CPU sessions with the same notebook code. The scored file of each method is fixed in Table 1; further kinds of output (for example MICA's initial C$\alpha$ model) were scored as well and are reported in the Appendix, but are not used in the main comparison. The analysis script reads the status tables of all sessions and the scores, takes the latest result of every method and entry (for MICA, only from the final docking set-up), and writes the tables, the figures and the numbers quoted in the text (as LaTeX macros), so that the numbers of the results are not typed in by hand; the few figures quoted from logs (for example the candidate atoms above) are marked as such. Its quality checks stop on a method that is unknown, an entry that is not in the frozen list, a score for a run that did not finish, and more than one primary model per method and entry; the check on entries that have both an *ok* and a *failed* result flags exactly the 3 MICA reruns above.

## 5 Evaluation metrics

### 5.1 Scoring protocol

Every model is compared with the deposited model of its entry (protein chains only, first model, first alternative location) with the same code and tools for every method. The protocol was written down on 24 October 2026, before any MICA model existed, and was not changed afterwards. A model with several chains, fragments or copies is scored as the method wrote it. Table 2 lists the measures: the six of the MICA paper (Gyawali et al., 2025) plus the C$\alpha$ RMSD and the residue coverage, which I had named in my email to Prof. Cheng, and a fragment-agnostic TM-score that I added to read the first one correctly.

**Table 2.** Measures of the accuracy of a model. The completion and the run time are defined with their results (Section 6). The TM-score is the primary measure (spec decision 7 and 8). $N_{\text{close}}$ and $N_{\text{far}}$ are the residues of the model that have a deposited C$\alpha$ within 3.0 Å or not.

| Measure | Definition | Tool |
|---|---|---|
| TM-score (primary) | Normalised by the length of the *deposited* model; multi-chain alignment with the chains paired one to one (`-mm 1 -ter 1`), so extra copies in the model are ignored. | US-align (Zhang et al., 2022; Zhang and Skolnick, 2004) |
| Aligned length, RMSD, sequence identity | Aligned C$\alpha$ pairs of the same run, their RMSD (Å), and the share of them with the same residue type. | US-align |
| TM-score, fragment-agnostic | All chains of each structure joined into one chain and aligned without regard to residue order (`-mm 5`); fragment order, chain breaks and extra copies do not matter. | US-align |
| C$\alpha$ match (%) | $100\,N_{\text{close}}/(N_{\text{close}}+N_{\text{far}})$: the share of the model's C$\alpha$ atoms that lie within 3.0 Å of a deposited C$\alpha$ (a precision). | `phenix.chain_comparison` (Liebschner et al., 2019) |
| Sequence match (%) | Share of the matched residues that have the residue type of the deposited residue. | `phenix.chain_comparison` |
| Coverage | Residues of the model divided by residues of the deposited model; above 1 means that the model has more residues than the deposit (extra copies or extra chains). | – |
| C$\alpha$ quality score (%) | C$\alpha$ match $\times$ coverage, as defined for the MICA paper and its predecessor; it equals the share of deposited residues found when $N_{\text{close}}+N_{\text{far}}$ equals the residues of the model, so built copies cancel out. | – |

**Reading the measures.** Before `chain_comparison` each model is moved onto the deposited model with the rotation of the `-mm 1` run, because a map can hold symmetry copies of the deposited chain and a method may build another copy; the same measures without this step are in the Appendix. Two properties of the measures matter for the comparison. First, US-align's multi-chain mode aligns only one fragment per deposited chain: a model cut into many pieces scores low although each piece is right (a synthetic test gave 0.14 for a perfect model cut into seven fragments). ModelAngelo writes its models as several chains (median 4, up to 67; the deposited models have 1 to 4), so its TM-score mixes accuracy and continuity; the fragment-agnostic TM-score separates them. Second, the C$\alpha$ match is a precision: a method that builds every copy it finds in the map (ModelAngelo, EMProt) is penalised for copies that the deposited model does not contain, although the deposited chain is built correctly. Conclusions therefore rest on the TM-score and the C$\alpha$ quality score, and the other measures are shown next to them with the coverage.

## 5.2 Statistics

Each method is compared with MICA on the entries where both have a scored model; the number of pairs is always given, and every difference is *method minus MICA*. The primary test is the two-sided Wilcoxon signed-rank test on the TM-score; the paired *t*-test is given because the MICA paper uses it. Confidence intervals (95 %) are percentile bootstrap intervals over entries (10 000 resamples, seed 2026) for means, medians and mean differences. *p*-values are adjusted by Holm's method over the four method pairs, separately for each measure and group; they are not adjusted across measures, so every measure except the TM-score is exploratory. Results are given for all entries and for each resolution stratum. Failed runs are not missing data: a sensitivity analysis scores them as zero on every measure and compares the means over all 20 entries. The planned subset of low-identity entries has too few entries to test and is described. Spearman rank correlations with resolution, size and novelty are descriptive. With 17 pairs or fewer, differences of a few hundredths in the TM-score cannot be told from noise, so the report puts effect sizes and intervals

**Table 3.** Completion. *Model scored*: the run ended with a model file that was scored. *Failed*: crash, kill by the system or timeout. The last two columns divide by the 20 entries and by the entries on which the method was run (all 20).

| Method | Entries | Model scored | Failed | No usable model | Not run | Completion, of 20 (%) | of attempted (%) |
|---|---|---|---|---|---|---|---|
| MICA | 20 | 17 | 3 | 0 | 0 | 85 | 85 |
| ModelAngelo | 20 | 19 | 1 | 0 | 0 | 95 | 95 |
| EModelX(+AF) | 20 | 18 | 2 | 0 | 0 | 90 | 90 |
| EMProt | 20 | 19 | 1 | 0 | 0 | 95 | 95 |
| E3-CryoFold | 20 | 15 | 5 | 0 | 0 | 75 | 75 |

**Table 4.** TM-score and three other measures of the scored models of each method (all entries). Medians for the C$\alpha$ match, coverage and C$\alpha$ quality; the other measures are in Table A12. $n$ is the number of scored models.

| Method | $n$ | TM-score, median (IQR) | TM-score, mean [95% CI] | C$\alpha$ match | Coverage | C$\alpha$ quality |
|---|---|---|---|---|---|---|
| MICA | 17 | 0.94 (0.91–0.98) | 0.94 [0.91, 0.96] | 94.9 | 1.02 | 96.0 |
| ModelAngelo | 19 | 0.91 (0.67–0.99) | 0.79 [0.68, 0.90] | 91.1 | 1.01 | 96.7 |
| EModelX(+AF) | 18 | 0.94 (0.91–0.98) | 0.94 [0.91, 0.96] | 94.1 | 1.01 | 94.7 |
| EMProt | 19 | 0.98 (0.96–0.99) | 0.96 [0.92, 0.98] | 57.4 | 1.45 | 98.2 |
| E3-CryoFold | 15 | 0.91 (0.77–0.97) | 0.86 [0.79, 0.93] | 80.9 | 1.03 | 92.0 |

before $p$-values.

# 6 Results

## 6.1 Completion

Table 3 shows how many of the 20 entries ended with a scored model. MICA built 17 (85 %), ModelAngelo 19 (95 %), EModelX(+AF) 18 (90 %), EMProt 19 (95 %) and E3-CryoFold 15 (75 %). No run produced a file that could not be scored. The 9VI4 entry (1.8 Å, a 1 GB map) failed for every method, and 26DK failed for MICA, EModelX(+AF) and E3-CryoFold (Table A8, Appendix). The three failures of MICA are limits of the free machine and not errors of the program: two kills by the system when MICA's model building used up the RAM, and one run that hit the 2-hour limit close to the end of its model building (the last gap of the last chain). They count as failures all the same, because the user who runs MICA on this machine does not get a model. Of the 17 models, 3 (22FX, 9WQ6 and 9V4D) needed a second attempt after a GPU out-of-memory failure of the first one (Section 4.4).

## 6.2 Accuracy of the models that were built

Table 4 and Figure 1 show the accuracy of each method on its own set of models; the sets differ in size, so the rows are not directly comparable. The median TM-score of MICA is 0.94 (interquartile range 0.91–0.98; mean 0.94, 95 % CI 0.91–0.96), and 15 of its 17 models reach 0.90 or more (the lowest is 0.80). EModelX(+AF) has the same median and mean (0.94, 0.94). EMProt is higher (median 0.98, mean 0.96). E3-CryoFold is lower (0.91, 0.86). ModelAngelo has a median of 0.91 but a mean of only 0.79, because 3 of its models are below 0.5 (9WQ6, 9VBD, 9WTX, all at resolutions above 3.0 Å).

## 6.3 Paired comparison with MICA

Table 5 and Figure 2 compare each method with MICA on the entries where both built a model (difference = method minus MICA).

- **EMProt** has a higher TM-score than MICA on 16 of 17 entries; the mean difference is 0.04 (95 % CI 0.02 to 0.06), the median difference 0.02, the Wilcoxon $p$ is $<0.001$ (Holm $<0.001$; paired $t$-test, Holm: 0.024). The difference is small and consistent: it is positive in both strata (0.05 up

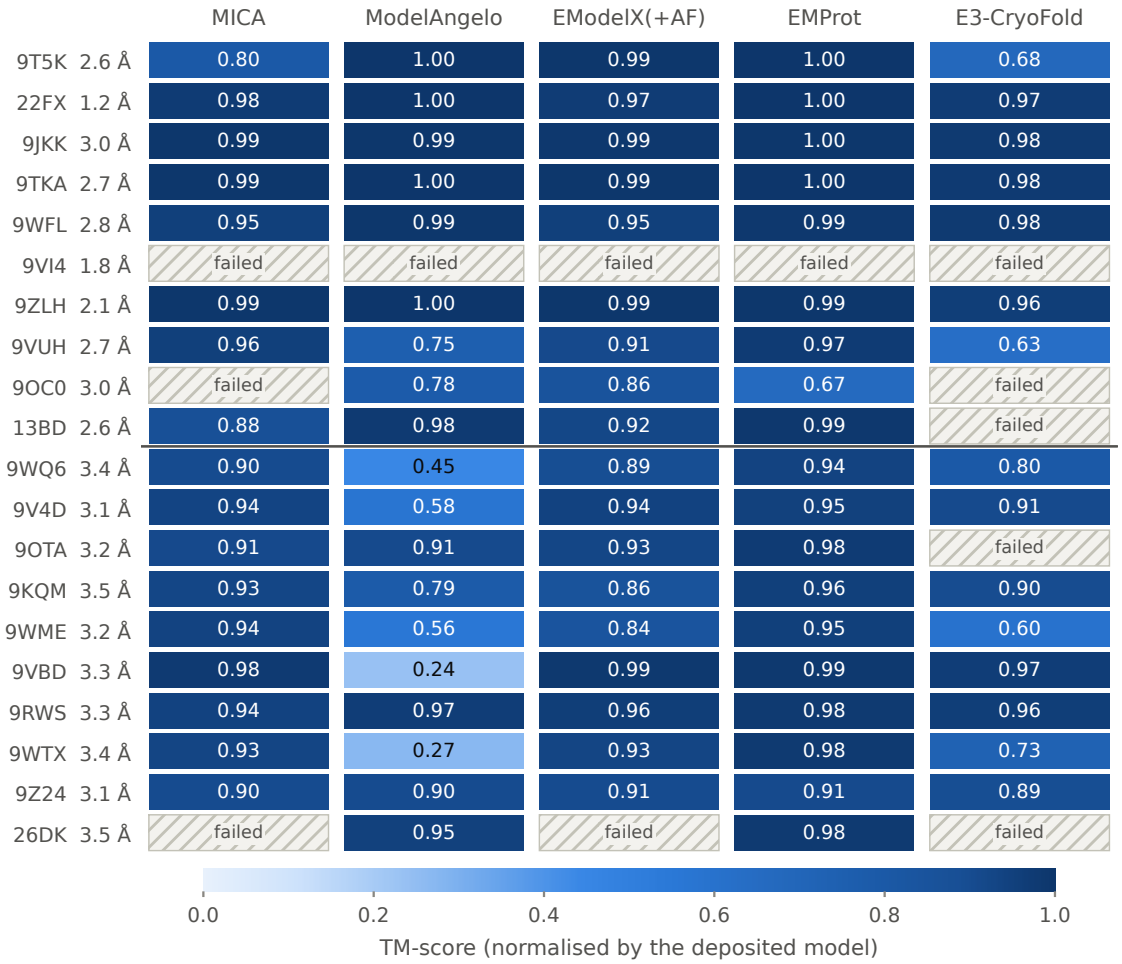



**Figure 1.** TM-score of every method on every entry (the primary measure, normalised by the deposited model). Entries are ordered by stratum, then by the seeded order of the selection; hatched cells are failed runs. The line separates the two resolution strata.

to 3.0 Å, 0.03 above) and EMProt is higher on 9 of the 9 entries above 3.0 Å.

- **EModelX(+AF)** is not different from MICA: mean difference 0.00 (−0.02 to 0.03), higher on 9 and lower on 8 of 17 entries, Wilcoxon Holm $p = 0.818$.
- **E3-CryoFold** is lower than MICA: −0.08 (−0.14 to −0.03) over 15 entries, lower on 13 of them; Wilcoxon Holm $p = 0.020$, but the $t$-test with Holm correction gives 0.067.
- **ModelAngelo** has a mean difference of −0.15 (−0.28 to −0.03) but a median difference of 0.00 (higher on 7, lower on 10 of 17 entries); the Wilcoxon Holm $p$ is 0.414 and the $t$-test Holm $p$ is 0.076. The mean is carried by the entries above 3.0 Å: up to 3.0 Å the mean difference is 0.02 (−0.06 to 0.09, $n = 8$); above 3.0 Å it is −0.30 (−0.48 to −0.13, $n = 9$), with ModelAngelo lower on 8 of them and a Wilcoxon Holm $p$ of 0.059 (0.041 for the $t$-test).

The rank correlation of the TM-score with the resolution is weak for MICA (Spearman $\rho = -0.25$, $p = 0.325$) and stronger for ModelAngelo ($\rho = -0.66$, $p = 0.002$), EMProt (−0.53, 0.018) and EModelX(+AF) (−0.48, 0.044); the correlations with the number of residues are all weak (Table A16, Appendix).

### 6.4 What the other measures say

The ranking by TM-score is not an artefact of fragmentation or of extra copies. ModelAngelo writes its models as several chains; US-align's multi-chain mode counts one fragment per deposited chain, so its TM-score mixes accuracy and continuity. With the fragment-agnostic TM-score (Table A14) the difference between ModelAngelo and MICA shrinks to −0.07 (−0.14 to −0.00; Wilcoxon

**Table 5.** Paired comparison of the TM-score with MICA on the entries where both methods built a model (method minus MICA; 95 % bootstrap CI of the mean difference; two-sided Wilcoxon signed-rank test and paired *t*-test, both with Holm correction over the four methods; counts of entries where the method is higher, lower or equal).

| Method | Pairs | Mean difference [95% CI] | Median diff. | Wilcoxon $p$ | Holm $p$ | $t$-test Holm $p$ | higher / lower / equal |
|---|---|---|---|---|---|---|---|
| *All entries* | | | | | | | |
| ModelAngelo | 17 | −0.15 [−0.28, −0.03] | 0.00 | 0.207 | 0.414 | 0.076 | 7 / 10 / 0 |
| EModelX(+AF) | 17 | 0.00 [−0.02, 0.03] | 0.00 | 0.818 | 0.818 | 0.827 | 9 / 8 / 0 |
| EMProt | 17 | 0.04 [0.02, 0.06] | 0.02 | <0.001 | <0.001 | 0.024 | 16 / 1 / 0 |
| E3-CryoFold | 15 | −0.08 [−0.14, −0.03] | −0.03 | 0.007 | 0.020 | 0.067 | 2 / 13 / 0 |
| *Resolution ≤ 3.0 Å* | | | | | | | |
| ModelAngelo | 8 | 0.02 [−0.06, 0.09] | 0.01 | 0.250 | 0.500 | 0.785 | 6 / 2 / 0 |
| EModelX(+AF) | 8 | 0.02 [−0.01, 0.08] | 0.00 | 1.000 | 1.000 | 0.785 | 3 / 5 / 0 |
| EMProt | 8 | 0.05 [0.01, 0.10] | 0.02 | 0.016 | 0.062 | 0.354 | 7 / 1 / 0 |
| E3-CryoFold | 7 | −0.07 [−0.16, −0.00] | −0.01 | 0.109 | 0.328 | 0.566 | 1 / 6 / 0 |
| *Resolution > 3.0 to 4.0 Å* | | | | | | | |
| ModelAngelo | 9 | −0.30 [−0.48, −0.13] | −0.36 | 0.020 | 0.059 | 0.041 | 1 / 8 / 0 |
| EModelX(+AF) | 9 | −0.01 [−0.04, 0.01] | 0.00 | 1.000 | 1.000 | 0.317 | 6 / 3 / 0 |
| EMProt | 9 | 0.03 [0.02, 0.04] | 0.03 | 0.004 | 0.016 | 0.011 | 9 / 0 / 0 |
| E3-CryoFold | 8 | −0.09 [−0.17, −0.02] | −0.03 | 0.039 | 0.078 | 0.168 | 1 / 7 / 0 |

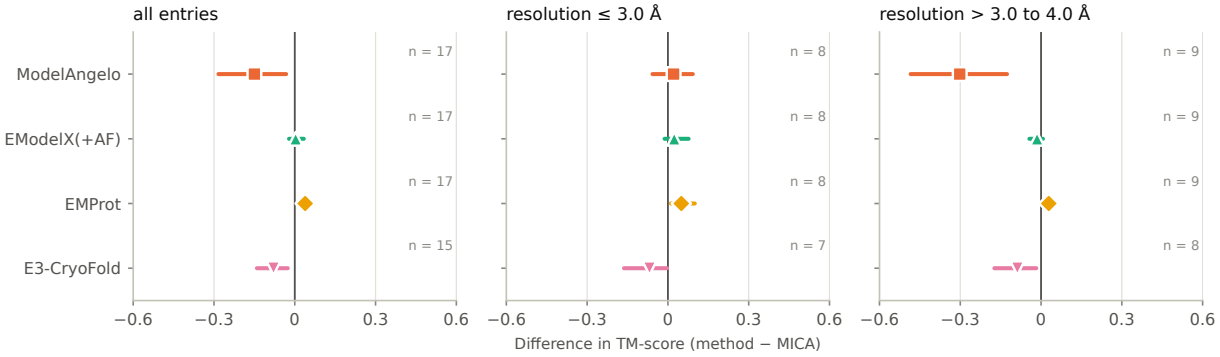



**Figure 2.** Difference in TM-score between each method and MICA (mean and 95 % bootstrap CI), for all entries and for each resolution stratum; positive values favour the other method.

Holm $p$ = 0.487), EModelX(+AF) stays equal (0.00), EMProt stays higher (0.03, Holm $p$ = 0.007) and E3-CryoFold lower (−0.05, Holm $p$ = 0.007). The Cα match (precision) is lowest for EMProt (median 57.4 against 94.9 for MICA) because EMProt builds 1.45 times the deposited number of residues (median; up to 25.11 times), that is, every copy it finds in the map; the same holds for ModelAngelo in a few entries (coverage up to 23.85). MICA builds about the deposited number of residues (median coverage 1.02) because its input is one AF3 prediction per unique chain; only in 13BD did it build clearly more than the deposit (coverage 4.76; the next highest is 1.18). The Cα quality score, which does not punish extra copies, gives 96.0 for MICA, 96.7 for ModelAngelo, 94.7 for EModelX(+AF), 98.2 for EMProt and 92.0 for E3-CryoFold (medians, %). The median sequence match of MICA is 98.5 %, and its median RMSD over the aligned residues is 1.10 Å for 409 aligned residues, against 0.63 Å for 306 for ModelAngelo, whose models align over fewer residues, so RMSDs are not comparable across methods (all measures: Table A12, Figure A1).

## 6.5 Failures scored as zero

If a failed run counts as zero on every measure (Table A9), the mean TM-score over all 20 entries is 0.80 for MICA (95 % CI 0.64–0.93), 0.76 for ModelAngelo, 0.84 for EModelX(+AF), 0.91 for EMProt and 0.65 for E3-CryoFold. On this footing neither ModelAngelo nor EModelX(+AF) differs from MICA: the mean differences are −0.04 and 0.05 (Holm $p$ = 1.000 and 1.000). EMProt remains higher (0.12; Holm $p$ <0.001) and E3-CryoFold lower (−0.15; Holm $p$ = 0.005). If the 3 reruns counted as failures as well, MICA would have 14 models with a mean TM-score of 0.94, and a mean of 0.66

over all 20 entries.

### 6.6 Docking and its effect on MICA

MICA's own docking could not be run on a free machine (Section 2.2), so the placements of the AF3 domains in the map are coarse and unverified. Pass 3 docked 46 of 59 domains (Table A6); 14 entries are fully docked, 5 partly and 1 not at all.

**Docked against no docked input.** On the 8 entries that ran with and without docked input (all fully docked), the TM-score is 0.953 with and 0.936 without docked input; the mean difference is 0.017 (95 % CI 0.002 to 0.037), Wilcoxon $p = 0.039$, higher with docked input on 6 and lower on 2 entries (Table A10). The differences are small (the largest are 9KQM and 9VUH), but they go in the expected direction, which is weak evidence that the coarse placements carry information. Without docked input MICA still uses the AF3 structures when it builds the chains, so this is not a comparison with MICA without AlphaFold.

**By docking mode.** The 13 models of fully docked entries have a mean TM-score of 0.95 (range 0.90–0.99), the 4 models of partly docked entries 0.89 (0.80–0.98); the three lowest TM-scores of MICA are those of 9T5K, 13BD and 9Z24, and 3 of these entries are partly docked. This is a description and not a test, because the entries also differ in size and resolution, but it points to the same side: where the free-compute docking left a domain unplaced, MICA was less accurate, so the docking deviation more probably understates than overstates MICA. The initial C$\alpha$ model that MICA's network writes before the chains are built from the AF3 structures has a median TM-score of 0.82, against 0.94 for the final model (Table A15); a large part of MICA's accuracy on these maps comes from the step that uses the AF3 structures (the AF3 predictions themselves were not scored).

### 6.7 Novelty and run time

Only 3 entries have a chain at 30 % identity or less to any pre-2026 PDB chain (9TKA, 9VBD, 26DK), and MICA failed on one of them, so the planned low-identity analysis has 2 pairs and is not tested; the TM-scores on these entries are in Table A11. Over all entries the TM-score of MICA is not related to the identity to older entries (Spearman $\rho = 0.02$, $p = 0.940$), and no method shows a relation with $p$ below 0.05 (smallest $p$ of the five methods: 0.080; Table A16).

Run times are only indicative (Table A17, Figure A4). The median time of MICA is 82 min per entry (range 33 to 558; 40.0 h in total for programs 1 to 3 including docking), against 20 min for ModelAngelo, 7 for EModelX(+AF), 14 for EMProt and 3 for E3-CryoFold. The baselines were timed while a second program used the other GPU, MICA's program 3 ran one entry at a time, and most of MICA's time on a large map goes into model building on one CPU core. The whole benchmark, including every failed attempt and the first two docking passes, used 36.4 h of GPU sessions and 68.0 h of CPU sessions (23 sessions, Table A5).

## 7 Discussion and limitations

### 7.1 What the results show

On maps released in 2026 and within the size range of this benchmark, MICA gives accurate models when it completes: the median TM-score is 0.94, and 15 of its 17 models reach 0.90 or more. With the same AF3 predictions, EModelX(+AF) is indistinguishable from it (mean difference 0.00); E3-CryoFold, which has no AF3 input, is less accurate ($-0.08$) and fails more often (5 of 20 entries). ModelAngelo, which also has no AF3 input, matches MICA on the maps up to 3.0 Å (0.02) and is lower above 3.0 Å ($-0.30$); part of this gap is an effect of how the TM-score treats fragments, because over all entries the mean difference shrinks from $-0.15$ to $-0.07$ with the fragment-agnostic score (Section 6.4). EMProt, which also uses the AF3 models but builds from the map first, is higher than MICA by a small and consistent margin (0.04); it appeared after MICA,

which may explain part of the gap, but this benchmark cannot separate the reasons. After the Holm correction only the difference to EMProt is clear on both tests (Wilcoxon $<0.001$, $t$-test 0.024); E3-CryoFold is lower on the Wilcoxon test (0.020) but not on the $t$-test (0.067), and ModelAngelo above 3.0 Å is borderline (0.059 and 0.041). The other differences are within the noise of 17 entries.

The step that uses the AF3 structures lifts the median TM-score from 0.82 (the C$\alpha$ model of the network) to 0.94 (Table A15); the AF3 predictions were not scored on their own, so how much of the result is the prior and how much is the map is not separated here. The prior is good: the median over the predictions of the mean pLDDT is 87.0, and for 13 of the 20 entries a chain at 95 % identity or more to an older PDB entry exists, which AF3 (whose training data end in 2021; Abramson et al., 2024) may have seen. The benchmark therefore measures how well a method turns a good prior and a new map into a model; the 3 entries with no relative in older PDB entries (9TKA, 9VBD, 26DK) are too few to say how MICA behaves without that prior (MICA failed on one of them; its scores on the others are in Table A11).

### 7.2 Limitations

1. **Sample and range.** There are 20 entries, 17 MICA models and at most 17 pairs per comparison, all single-particle, protein-only maps of up to 682 residues and three unique chains (the test maps of the MICA paper go up to several thousand residues). Differences of a few hundredths of TM-score cannot be resolved. Nothing here says how the methods behave on large complexes, where the memory and time limits of the free machine, which already stopped 2 MICA runs for lack of RAM and 1 at its time limit here, would matter even more.
2. **Docking.** MICA's own docking (Phenix molecular replacement without time limit) could not run on a free machine. The search here is a coarse Fourier search at 6 or 10 Å with a time limit; 5 entries are partly docked and 1 not at all, and the placements were not checked against the deposited models. The indirect evidence (Section 6.6) suggests that this understates MICA, but it rests on 8 entries and on a descriptive split.
3. **No real-space refinement.** MICA's last step stopped with a parameter error in Phenix 2.2.1-6174, for every entry. The scored models are MICA's models before refinement. I did not refine the outputs of the other methods either, but their own pipelines may include refinement-like steps, so the comparison is not strictly symmetric; the absolute numbers are not those of the authors' full pipeline, and the effect on all-atom quality was not measured.
4. **Memory patch and reruns.** The network of MICA predicts every window on its own, as MICA does for maps with 200 windows or fewer, so that the larger maps fit into the 15 GB of the GPU; MICA's own setting (batches of eight windows) was not compared with it, and the setting is meant to change memory use only. The 3 models of 22FX, 9WQ6 and 9V4D come from reruns after a first failure with MICA's default batching, an exception to the one-run rule that is reported.
5. **Failures depend on the machine.** Of the 12 runs that ended without a model, 7 were kills by the system (probably for lack of RAM), 2 reached a run limit (1 of MICA and 1 of ModelAngelo) and 3 were crashes of E3-CryoFold (exit code 1; entries 13BD, 9OC0 and 9OTA). All happened on a machine with 31 GB of RAM, a 15 GB GPU and fixed run limits; a larger machine would probably have completed the kills and perhaps the runs that reached a limit, but not the crashes. Scoring failures as zero therefore describes this machine and the programs together.
6. **One run per entry.** Each method ran once per entry, with one AF3 prediction (one random seed, server defaults) per chain; the variation between AF3 seeds and between repeated runs was not measured. Run times come from shared free machines and are indicative.
7. **Reference models and measures.** The deposited models are not ground truth: they contain errors and may have been built with tools such as ModelAngelo or from AF predictions, which could favour similar methods. The maps often hold more copies than the deposit, so methods that build every copy have a low C$\alpha$ match by construction; US-align's TM-score counts one fragment per chain. The conclusions rest on the TM-score and the C$\alpha$ quality score and are checked with the fragment-agnostic score.

8. **Protocol choices.** EModelX(+AF) was run with the AF3 predictions as templates, not in its default AlphaFold mode, and E3-CryoFold only in its `pre_align` protocol; ModelAngelo had no AlphaFold input by design. These choices were fixed before the runs. The newer methods are two, because CryoAtom could not run within the compute budget, and the literature search could not use forward citations of MICA.
9. **Selection.** The map–model correlation rule ($CC_{mask}$ of at least 0.70) removed 6 entries, all in the stratum above 3.0 Å, and 2 more could not be computed; the stratum above 3.0 Å therefore holds maps that are well explained by their models. The seeded order favours common specimens to the degree that the pool does.

## 8 Conclusions

1. On 20 maps released in 2026, MICA built an accurate model (median TM-score 0.94) for 17 entries and did not finish on 3: two runs were killed by the system for lack of RAM and one reached its time limit, all on a free 31 GB machine.
2. Where both methods built a model (differences are method minus MICA), EModelX(+AF) did not differ from MICA with the same AF3 predictions (0.00), EMProt was slightly higher (0.04; Holm $p < 0.001$), E3-CryoFold was lower ($-0.08$; Holm $p = 0.020$ for the Wilcoxon test, 0.067 for the $t$-test), and ModelAngelo was lower on the maps above 3.0 Å ($-0.30$; Holm $p = 0.059$ and 0.041) but not up to 3.0 Å (0.02). Only the difference to EMProt is clear on both tests.
3. Counting every failure as zero, the mean TM-score over all 20 entries is 0.80 for MICA, 0.76 for ModelAngelo, 0.84 for EModelX(+AF), 0.91 for EMProt and 0.65 for E3-CryoFold; MICA does not differ from ModelAngelo or EModelX(+AF) on this footing, is below EMProt and above E3-CryoFold.
4. The step of MICA that uses the AF3 structures raises the median TM-score from 0.82 (the network's own C$\alpha$ model) to 0.94. Docked input helped slightly on the 8 entries that could be tested (mean difference 0.017).
5. The results hold for the pipeline as I could run it: coarse docking, no real-space refinement, a one-window memory patch, one run per entry. What would tighten them: MICA's own docking on a large-memory machine, its refinement with a Phenix version that accepts its parameter file, a check of the docked placements against the deposited models, repeated AF3 seeds and larger entries.

## 9 Reproducibility and statement on assistance

**Repository.** https://github.com/Zohaib-Bioinfo/mica-benchmark holds the specification (`docs/SPEC.md`) and the project log with every decision, error and change (`LOG.md`); the frozen entry list and the selection record (`data/`); the notebooks as run, their generators and the analysis scripts with their tests (`scripts/`); the environment files (`env/`) and the pinned methods table (`methods/`); the unpacked text outputs of every session and the scores of every model (`results/raw/`); and this report (`report/`).

**Regenerating the report.** One script turns the scores and the status tables into the statistics (`analyze_results.py`), one draws the figures (`make_figures.py`), one writes the tables and the numbers of the text (`make_report_tables.py`) and one compiles the PDF (`build_report.py`), which stops if a number, a reference or a citation is missing. `scripts/regenerate.sh` runs the four in order (`--check`: compare the tables only); see the `README.md`.

**What is not in the repository.** The AF3 predictions (the terms of AlphaFold Server), the Phenix installer and the coordinate files of the models; the AF3 job list, the checks and the file hashes are public, so that the inputs can be recreated. Running the notebooks again needs a Kaggle account with GPU quota, the private AF3 dataset and a Phenix download login.

**Statement on AI assistance.** I did this project with an AI assistant, Claude (Anthropic), and the work was divided as follows. *My part:* I took the task from Prof. Cheng, chose the free-compute plan and locked the specification (`docs/SPEC.md`: the methods, the entry filters and the selection rule, the metrics, the statistics and the handling of failed runs) on 10 October 2026, before any method was run; I made the AF3 predictions, obtained the Phenix academic licence, ran every notebook that produced a result on my own Kaggle and Colab accounts, uploaded the outputs of each session and read this report in full. *The assistant's part:* it wrote the notebooks, the scripts and their tests, the project log and the text of this report, working from the specification and from the outputs that I uploaded; it proposed the plan for each stage and the technical decisions, which I approved. *Checks:* every result in the report is computed from the raw outputs by the scripts of the repository and inserted automatically (the few figures quoted from logs are marked as such, Section 4), and every decision, error and change, including the assistant's own mistakes (for example a Phenix parameter in the second docking pass that this Phenix version rejected), is dated in `LOG.md`. I am responsible for what this report says.

**Acknowledgements.** MICA, ModelAngelo, EModelX, EMProt and E3-CryoFold are the work of their authors; I used their public code and weights as released. Compute came from the free tiers of Kaggle and Google Colab.

## References

J. Abramson, J. Adler, J. Dunger, et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630:493–500, 2024. doi: 10.1038/s41586-024-07487-w.

P. V. Afonine et al. Real-space refinement in PHENIX for cryo-EM and crystallography. *Acta Crystallographica Section D*, 74:531–544, 2018.

S. Chen, S. Zhang, X. Fang, L. Lin, H. Zhao, and Y. Yang. Protein complex structure modeling by cross-modal alignment between cryo-EM maps and protein sequences. *Nature Communications*, 15:8808, 2024. doi: 10.1038/s41467-024-53116-5.

R. Gyawali, A. Dhakal, and J. Cheng. Multimodal deep learning integration of cryo-EM and AlphaFold3 for high-accuracy protein structure determination. *bioRxiv*, 2025. doi: 10.1101/2025.07.03.663071. Published in Communications Chemistry 8, 320 (2025), doi:10.1038/s42004-025-01718-5.

K. Jamali, L. Käll, R. Zhang, A. Brown, D. Kimanius, and S. H. W. Scheres. Automated model building and protein identification in cryo-EM maps. *Nature*, 628:450–457, 2024. doi: 10.1038/s41586-024-07215-4.

A. M. Lau, N. Bordin, S. M. Kandathil, I. Sillitoe, V. P. Waman, J. Wells, C. A. Orengo, and D. T. Jones. Merizo: a rapid and accurate protein domain segmentation method using invariant point attention. *Nature Communications*, 14:8445, 2023.

T. Li, J. Chen, H. Li, H. Cao, and S.-Y. Huang. EMProt improves structure determination from cryo-EM maps. *Nature Structural & Molecular Biology*, 33:341–350, 2026. doi: 10.1038/s41594-025-01723-1.

D. Liebschner et al. Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. *Acta Crystallographica Section D*, 75:861–877, 2019.

P. Rotkiewicz and J. Skolnick. Fast procedure for reconstruction of full-atom protein models from reduced representations. *Journal of Computational Chemistry*, 29:1460–1465, 2008.

M. Steinegger and J. Söding. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35:1026–1028, 2017.

Su et al. CryoAtom improves model building for cryo-EM. *Nature Structural & Molecular Biology*, 33:351–361, 2026. doi: 10.1038/s41594-025-01713-3.

J. Wang, C. Tan, Z. Gao, G. Zhang, Y. Zhang, and S. Z. Li. End-to-end cryo-EM complex structure determination with high accuracy and ultra-fast speed. *Nature Machine Intelligence*, 7:1091–1103, 2025. doi: 10.1038/s42256-025-01056-0.

C. Zhang, M. Shine, A. M. Pyle, and Y. Zhang. US-align: universal structure alignments of proteins, nucleic acids, and macromolecular complexes. *Nature Methods*, 19:1109–1115, 2022.

Y. Zhang and J. Skolnick. Scoring function for automated assessment of protein structure template quality. *Proteins*, 57:702–710, 2004.

# A Tables and figures that the text refers to

This appendix holds the longer tables and the further figures. Every one of them is written by a script of the repository from the unpacked session outputs in `results/raw/` (Section 9); the tables are `report/generated/tab_*.tex` and the figures `report/figures/`. Tables and figures of the appendix are numbered A1, A2, ...

## A.1 Methods and entries

Table A1 lists every method that the screen of the newer methods considered, Table A2 the filters that led to the pool of candidates, Table A3 the twenty entries and Table A4 the pinned versions.

**Table A1.** Screen of newer methods (spec decision 2). *Included*: *baseline*, the method under test or one of the two baselines of its paper; *passed*, passed the screen (CryoAtom was then dropped by the first scope cut, Section 2.4); *reserve*, kept in reserve and not used; *no*, excluded for the reason given. The reasons are the notes written at the time of the search (to 11 October 2026).

| Method | Year | Included | Reason |
|---|---|---|---|
| MICA | 2025 | baseline | Method under test. |
| ModelAngelo | 2024 | baseline | One of the two methods tested in the MICA paper. |
| EModelX(+AF) | 2024 | baseline | The other method tested in the MICA paper. |
| CryoAtom | 2025 | passed | Newer than both baselines; deep learning; map + sequence to atomic model; public code and weights (MIT per README; the paper states no licence); defaults in one documented command; fits 16 GB on paper. Formerly CryFold. The CryoAtom2 preprint (proteins and nucleic acids) reports nucleic-acid results in its abstract; whether protein-only accuracy changed from v1 is not stated. |
| EMProt | 2025 | passed | Newer than both baselines; deep learning; map + sequence to atomic model; accepts per-chain predicted models, so it can take the same AF3 inputs as MICA; public code and weights; fits 16 GB on paper. Licence wording differs: README says MIT, the paper says free for academic or noncommercial users. Read the LICENSE file before release. |
| E3-CryoFold | 2025 | passed | Newer than both baselines; deep learning; map + sequence to atomic model; public code and weights (MIT per README; the paper states no licence); fits 16 GB on paper. Defaults to be taken verbatim from the README example. First excluded by mistake on an inferred link to EModelX; corrected 2026-10-11 after reading its own paper. |
| EM3DFold | 2026 | reserve | First reserve. Newer, deep learning, map + sequence to atomic model (protein-only mode available), public code and downloadable weights, fits 16 GB on paper; licence not stated. A preprint ('in submission'), so under the tie-break the three peer-reviewed methods keep their places. From the same group as EMProt; the README does not say whether it supersedes EMProt. Training-data cutoff not read: check it before use, because the benchmark relies on the 2026 entries being unseen. Recheck on 2026-10-15 for a release tag, a licence and journal status. |
| CryoZeta | 2026 | reserve | Newer, deep learning, public code and weights, but fails the 16 GB rule on paper (README: 32 GB or more) and needs MSAs from an outside tool. Second reserve, after EM3DFold: used only if EM3DFold has already replaced a method or cannot be run, and only after one pilot on the smallest entry shows it fits 16 GB. |

| Method | Year | Included | Reason |
|---|---|---|---|
| CryoACE | 2026 | no | Fails the 16 GB rule (~38 GB) and is not a map-plus-sequence builder with documented defaults: it needs a rough initial model aligned to each map plus resolution and threshold settings. Its paper compares against ModelAngelo, CryoAtom, E3-CryoFold, Boltz-1 and CryoBoltz. |
| CryoBoltz | 2025 | no | Guided sampling of a Boltz model toward the map, with a pre-aligned starting model and per-map settings, not a map-plus-sequence builder with documented defaults. Also over 16 GB on the third-party figure. |
| CryoForge | 2026 | no | A post-builder correction layer that repairs another builder's model; it is not a map-plus-sequence builder. No code or weights stated. |
| CryoNet.Discovery | 2026 | no | Identifies known structures by matching map and structure embeddings (retrieval), so it does not build a model. No code or weights stated. Compared with ModelAngelo, cryoID and DomainFit. |
| CoCoFold | 2026 | no | Not a map-based builder (the input is raw particle images) and peak GPU memory is far above 16 GB. Its paper compares against DiffModeler, DeepMainmast, EModelX, MICA and ModelAngelo. |
| Cryo2Struct | 2024 | no | Not newer: published before EModelX (Nat. Commun. 15, 8808, same volume). Earlier method from the Cheng lab (Giri and Cheng); its evaluation code is checked in Phase 1 for the metric definitions. |
| DeepMainmast | – | no | Not newer: first published online in 2023, before the ModelAngelo paper (2024). |
| EMBuild | 2022 | no | Not newer (2022). |
| DiffModeler | 2024 | no | Built for low-resolution maps; outside the <=4.0 A test range. |
| DMcloud | 2025 | no | Built for medium- to low-resolution maps; outside the <=4.0 A test range. |
| DeepTracer (family) | – | no | Earlier family of methods; no 2025-2026 release found. |
| CryoREAD | – | no | Models DNA and RNA only; the benchmark is protein-only. |
| SMARTFold | 2023 | no | Not newer: 2023 preprint. |
| CryoNet.Refine | 2026 | no | Refines an existing model; does not build one from a map and sequence. |
| Cryosearch | 2026 | no | Assembles components into complexes rather than building from a map and sequence; no public code found. |
| CryoSampler | 2026 | no | Models conformational ensembles; no public code found. |
| Cryformer | – | no | Listed as its own entry in a bioRxiv review of Dec 2025 (10.64898/2025.12.28.696775; map + ESM-2 embeddings), but the name matches the encoder module of the CryFold/CryoAtom network (Cry-Net), and no separate paper or code was found. |
| SEGEM++ | – | no | Listed in the Dec 2025 review as using AlphaFold2 output; no paper or code found under this name. |
| Cryo2Struct2 | – | no | Described in the Dec 2025 review as map + ESM embeddings with AlphaFold3 refining its output; no paper, code or weights found under this name. How it relates to MICA is not established. |
| CryoCodex | 2026 | no | Screened by title only: map post-processing, not model building. |
| EMAlign | 2026 | no | Screened by title only: aligns cryo-EM maps, does not build models. |
| HDOCK-Multimer | 2026 | no | Screened by title only: predicts large complexes by docking and assembly; the title names no map input. |
| Ligand-modelling paper | – | no | Screened by title only: models ligands, not protein chains. |

**Table A2.** From the pool of cryo-EM entries released from 1 January to 12 October 2026 to the candidates. The map–model correlation rule and the clustering (Section 3) are applied to the 484 candidates of the last row.

| Step | Entries left | Removed |
|---|---|---|
| Cryo-EM entries released in the window (RCSB search) | 6 116 | – |
| mmCIF file of at most 40 MB compressed (a larger file holds far more than 700 residues) | 6 115 | 1 |
| PDB model released inside the window | 6 115 | 0 |
| Single-particle reconstruction | 5 700 | 415 |
| Resolution at most 4.0 Å | 5 219 | 481 |
| Protein only (every polymer entity is polypeptide(L)) | 3 977 | 1 242 |
| 1 to 700 modeled residues in total | 567 | 3 410 |
| At most 3 unique chain sequences | 548 | 19 |
| At most 5% UNK residues | 548 | 0 |
| Exactly one EMDB map linked to the entry | 485 | 63 |
| EMDB map released inside the window (EMDB API) | 484 | 1 |

**Table A3.** The twenty entries. *Residues*: modelled residues of the deposit. *Unique / all chains*: unique chain sequences and chains of the deposit. *CC_mask*: map–model correlation (Phenix `map_model_cc`, at the resolution of the entry). *Max. identity*: highest sequence identity of any chain to a chain of a PDB entry released before 2026 (MMseqs2; 0 = no hit). *AF3 pLDDT*: mean pLDDT of the AF3 predictions of the entry, weighted by residues. *Map*: size of the EMDB map file.

| PDB | EMDB | Res. (Å) | Residues | Unique / all chains | CC_mask | Max. identity to older PDB | AF3 pLDDT | Map (MB) |
|---|---|---|---|---|---|---|---|---|
| *Resolution ≤ 3.0 Å* | | | | | | | | |
| 9T5K | 55586 | 2.60 | 381 | 1 / 1 | 0.82 | 0.61 | 85 | 250 |
| 22FX | 68251 | 1.24 | 173 | 1 / 1 | 0.74 | 1.00 | 97 | 75 |
| 9JKK | 61564 | 3.00 | 530 | 1 / 1 | 0.84 | 1.00 | 85 | 66 |
| 9TKA | 56022 | 2.70 | 247 | 1 / 1 | 0.76 | 0.00 | 88 | 182 |
| 9WFL | 65930 | 2.75 | 682 | 2 / 2 | 0.84 | 1.00 | 91 | 66 |
| 9VI4 | 65077 | 1.80 | 516 | 1 / 1 | 0.87 | 0.98 | 98 | 1024 |
| 9ZLH | 74403 | 2.14 | 206 | 1 / 1 | 0.85 | 1.00 | 92 | 66 |
| 9VUH | 65363 | 2.70 | 634 | 3 / 3 | 0.74 | 1.00 | 77 | 128 |
| 9OC0 | 70305 | 2.97 | 512 | 2 / 4 | 0.83 | 1.00 | 84 | 333 |
| 13BD | 76936 | 2.65 | 461 | 1 / 2 | 0.88 | 0.95 | 84 | 281 |
| *Resolution > 3.0 to 4.0 Å* | | | | | | | | |
| 9WQ6 | 66160 | 3.40 | 261 | 1 / 1 | 0.76 | 1.00 | 84 | 128 |
| 9V4D | 64777 | 3.07 | 276 | 2 / 2 | 0.82 | 0.50 | 71 | 54 |
| 9OTA | 70828 | 3.20 | 534 | 1 / 2 | 0.85 | 1.00 | 86 | 69 |
| 9KQM | 62507 | 3.50 | 403 | 1 / 1 | 0.79 | 1.00 | 76 | 66 |
| 9WME | 66088 | 3.20 | 446 | 2 / 2 | 0.70 | 1.00 | 90 | 23 |
| 9VBD | 64920 | 3.26 | 521 | 1 / 1 | 0.77 | 0.00 | 93 | 128 |
| 9RWS | 54346 | 3.28 | 512 | 1 / 1 | 0.85 | 0.44 | 88 | 128 |
| 9WTX | 66226 | 3.37 | 500 | 1 / 1 | 0.72 | 0.76 | 88 | 128 |
| 9Z24 | 73762 | 3.10 | 338 | 1 / 1 | 0.87 | 1.00 | 80 | 221 |
| 26DK | 80558 | 3.50 | 191 | 1 / 1 | 0.74 | 0.00 | 97 | 16 |

| Method | Year | Included | Reason |
|---|---|---|---|
| TomoSwin3D | – | no | Screened by title only: identifies and classifies macromolecules in cryo-ET tomograms, does not build models. |

## A.2 Sessions, docking and failures

Table A5 lists the sessions, Table A6 the docking of every entry in the three passes, Table A7 MICA's outcome by entry and Table A8 every failed run.

**Table A4.** Pinned versions of the methods and the tools, as recorded in the run information of the sessions.

| Component | Commit or file (pinned) |
|---|---|
| MICA | `726d1bb14be04887a43b7f5aefcff0d843761d01` |
| ModelAngelo | `994945bdfa6e5368e0d62349a47792f4864eebc3` |
| EModelX(+AF) | `53713cf820d10d78b1ca6669ef197850cf907843` |
| EMProt | `9489d52e7fdc81c85ba82e386abcfa05bac2b9d1` |
| E3-CryoFold | `241cef48da83b96519e62ab02cccd9db96d8e68b` |
| Phenix | `Phenix-2.2.1-6174-Linux-x86_64.sh` |
| US-align | `746ad8a2a4dea9f317075cc482df0ecb5440bdea` |

**Table A5.** The 23 Kaggle sessions that produced results (CPU sessions have no GPU). Status of the entries of a session: *ok*, finished with its output; *docked*, programs 1 and 2 of MICA finished and the docked state was saved for a GPU session; *failed*, ended without the output of its step; *deferred*, program 3 of MICA was switched off after two failures in a row and the entry waits; *interrupted*, cut by the end of the session; *killed_corun*, a baseline killed by the system while the other GPU was busy (it ran again alone, Section 4); *not run*, no work was started. The baseline sessions count runs (method × entry), the MICA sessions count entries, and the scoring sessions the models that they scored. The result of `mica_cpu_c` for 9VUH (pass 1) is not used: every MICA result comes from the pass-3 set-up.

| Notebook | Started (UTC) | Minutes | GPU | Entries by status, or models scored |
|---|---|---|---|---|
| `gpu_batch1` | 2026-10-23 05:35 | 521 | yes | ok 64, failed 3, killed_corun 6, not run 7 |
| `mica_cpu_a` | 2026-10-24 06:34 | 417 | no | failed 3, deferred 4 |
| `gpu_batch2` | 2026-10-24 06:44 | 410 | yes | ok 1, failed 4, interrupted 1, not run 7 |
| `mica_cpu_c` | 2026-10-24 06:40 | 441 | no | ok 1, failed 4, deferred 1 |
| `mica_cpu_b` | 2026-10-24 06:37 | 611 | no | failed 5, deferred 2 |
| `mica_dock2_a` | 2026-10-25 16:17 | 262 | no | docked 4, failed 1 |
| `mica_dock2_d` | 2026-10-25 16:23 | 284 | no | docked 2, failed 3 |
| `mica_dock2_b` | 2026-10-25 16:19 | 377 | no | failed 5 |
| `mica_dock2_c` | 2026-10-25 16:21 | 379 | no | docked 1, failed 4 |
| `gpu_batch3` | 2026-10-25 16:30 | 432 | yes | ok 6, failed 2 |
| `mica_dock3_c` | 2026-10-26 10:46 | 229 | no | docked 5 |
| `mica_dock3_a` | 2026-10-26 10:42 | 272 | no | docked 5 |
| `mica_dock3_d` | 2026-10-26 10:48 | 275 | no | docked 5 |
| `mica_dock3_b` | 2026-10-26 10:44 | 459 | no | docked 5 |
| `scoring_base` | 2026-10-27 02:08 | 26 | no | 127 models scored |
| `mica_gpu_nodock` | 2026-10-27 02:05 | 105 | yes | ok 8, failed 3, deferred 9 |
| `mica_gpu` | 2026-10-27 02:04 | 143 | yes | ok 8, failed 3, deferred 9 |
| `scoring_mica` | 2026-10-27 10:15 | 14 | no | 24 models scored |
| `scoring_nodock` | 2026-10-27 10:18 | 14 | no | 24 models scored |
| `mica_gpu2_b` | 2026-10-27 16:38 | 190 | yes | ok 6, failed 1 |
| `mica_gpu2_a` | 2026-10-27 16:34 | 195 | yes | ok 3, interrupted 2 (cut by the time budget) |
| `scoring_mica2` | 2026-10-28 05:45 | 17 | no | 27 models scored |
| `mica_gpu3` | 2026-10-28 05:02 | 186 | yes | failed 2 |

**Table A6.** Docking of the AF3 domains in the three passes (Section 4.4): domains docked of the domains found by Merizo, and the median local correlation (CC) of the docked domains. The correlations of pass 1 and pass 3 are not comparable (pass 3 is measured at 6 Å after a coarse refinement). *Mode*: docking mode of the entry in pass 3. The last row sums the domains of the 20 entries; its CC is the median over all docked domains of the pass.

| Entry | Domains | Pass 1 docked | CC | Pass 2 docked | CC | Pass 3 docked | CC | Mode |
|---|---|---|---|---|---|---|---|---|
| 9T5K | 2 | 1 / 2 | 0.47 | 1 / 2 | 0.47 | 1 / 2 | 0.11 | partial |
| 22FX | 2 | 0 / 2 | – | 0 / 2 | – | 1 / 2 | 0.09 | partial |
| 9JKK | 2 | 0 / 2 | – | 0 / 2 | – | 2 / 2 | 0.11 | full |
| 9TKA | 1 | 1 / 1 | 0.44 | 1 / 1 | 0.44 | 1 / 1 | 0.08 | full |
| 9WFL | 5 | 1 / 5 | 0.47 | 1 / 5 | 0.47 | 5 / 5 | 0.06 | full |
| 9VI4 | 3 | 0 / 3 | – | 0 / 3 | – | 0 / 3 | – | none |
| 9ZLH | 2 | 0 / 2 | – | 0 / 2 | – | 2 / 2 | 0.10 | full |
| 9VUH | 4 | 4 / 4 | 0.42 | 4 / 4 | 0.42 | 4 / 4 | 0.17 | full |
| 9OC0 | 7 | 0 / 7 | – | 0 / 7 | – | 3 / 7 | 0.05 | partial |
| 13BD | 3 | 2 / 3 | 0.21 | 0 / 3 | – | 1 / 3 | 0.06 | partial |
| 9WQ6 | 1 | 0 / 1 | – | 0 / 1 | – | 1 / 1 | 0.03 | full |
| 9V4D | 1 | 0 / 1 | – | 0 / 1 | – | 1 / 1 | 0.09 | full |
| 9OTA | 2 | 0 / 2 | – | 0 / 2 | – | 2 / 2 | 0.10 | full |
| 9KQM | 2 | 1 / 2 | 0.08 | 1 / 2 | 0.08 | 2 / 2 | 0.07 | full |
| 9WME | 7 | 2 / 7 | 0.46 | 2 / 7 | 0.46 | 7 / 7 | 0.08 | full |
| 9VBD | 4 | 0 / 4 | – | 0 / 4 | – | 4 / 4 | 0.11 | full |
| 9RWS | 3 | 0 / 3 | – | 0 / 3 | – | 3 / 3 | 0.09 | full |
| 9WTX | 3 | 0 / 3 | – | 0 / 3 | – | 3 / 3 | 0.10 | full |
| 9Z24 | 3 | 0 / 3 | – | 0 / 3 | – | 1 / 3 | 0.14 | partial |
| 26DK | 2 | 2 / 2 | 0.53 | 2 / 2 | 0.53 | 2 / 2 | 0.27 | full |
| All | 59 | 14 / 59 | 0.44 | 12 / 59 | 0.45 | 46 / 59 | 0.09 | |

### A.3 Further results

Tables A9 and A10 belong to Sections 6 (failures as zero) and 6.6 (docked input); the other tables give the TM-score of every method on every entry (Table A11), all measures (Tables A12, A13 and A14), the other output files of the methods (Table A15), the correlations (Table A16), the run times (Table A17) and the comparison with MICA without docked input (Table A18).

### A.4 Further figures

The figures show the distributions of the measures (Figures A1 and A5), the pairs of TM-scores (Figure A2), the outcomes and the run times (Figures A3 and A4), and the checks with the fragment-agnostic score and the resolution (Figures A6, A7 and A8).

**Table A7.** MICA by entry: domains, domains docked in pass 3, docking mode, outcome of program 3 (*failed → ok*: the first attempt failed with a GPU out-of-memory error and the rerun with the memory patch built the model) and TM-score of the model (F: no model).

| Entry | Domains | Docked | Mode | Program 3 | TM-score |
|---|---|---|---|---|---|
| 9T5K | 2 | 1 | partial | ok | 0.80 |
| 22FX | 2 | 1 | partial | failed → ok | 0.98 |
| 9JKK | 2 | 2 | full | ok | 0.99 |
| 9TKA | 1 | 1 | full | ok | 0.99 |
| 9WFL | 5 | 5 | full | ok | 0.95 |
| 9VI4 | 3 | 0 | none | failed | F |
| 9ZLH | 2 | 2 | full | ok | 0.99 |
| 9VUH | 4 | 4 | full | ok | 0.96 |
| 9OC0 | 7 | 3 | partial | failed | F |
| 13BD | 3 | 1 | partial | ok | 0.88 |
| 9WQ6 | 1 | 1 | full | failed → ok | 0.90 |
| 9V4D | 1 | 1 | full | failed → ok | 0.94 |
| 9OTA | 2 | 2 | full | ok | 0.91 |
| 9KQM | 2 | 2 | full | ok | 0.93 |
| 9WME | 7 | 7 | full | ok | 0.94 |
| 9VBD | 4 | 4 | full | ok | 0.98 |
| 9RWS | 3 | 3 | full | ok | 0.94 |
| 9WTX | 3 | 3 | full | ok | 0.93 |
| 9Z24 | 3 | 1 | partial | ok | 0.90 |
| 26DK | 2 | 2 | full | failed | F |

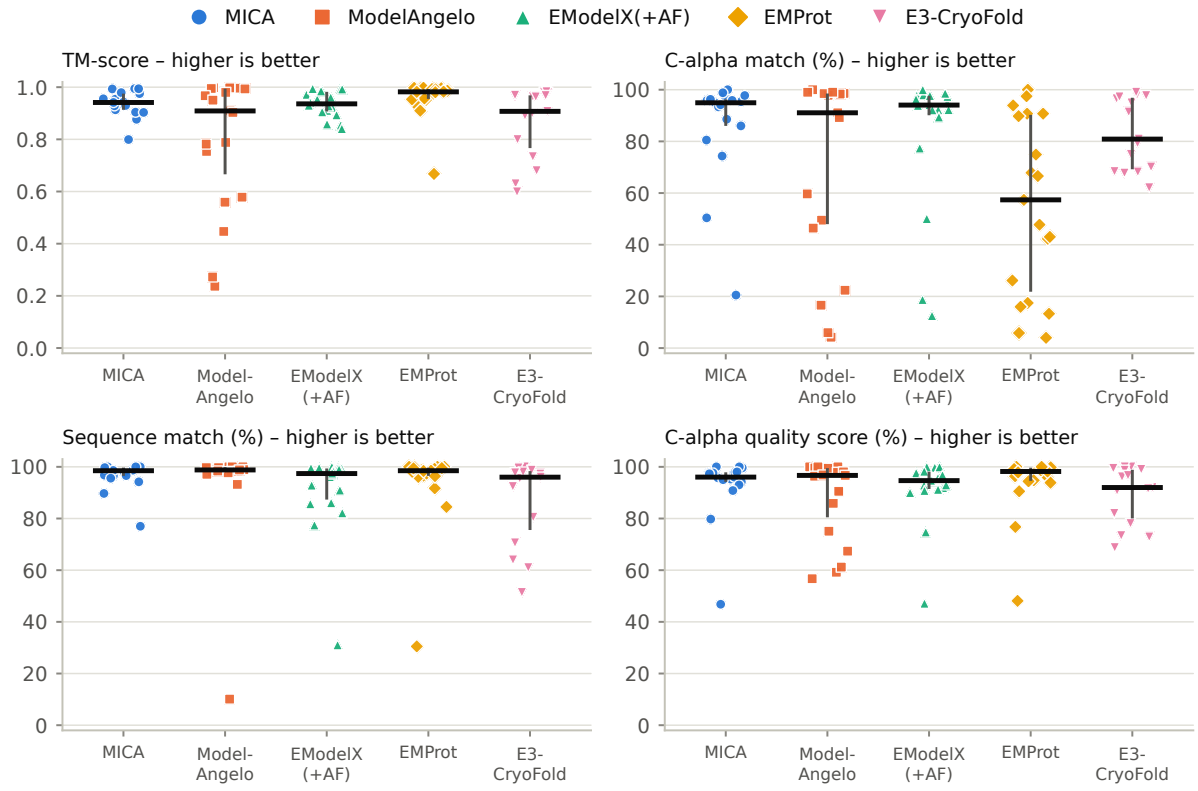



**Figure A1.** Four measures over the scored models of each method (one dot per entry; black bar: median; grey bar: interquartile range).

**Table A8.** Every run that ended without a scored model, with the reason recorded by the notebook. Exit code −9 is a kill by the system.

| Entry | Method | Outcome | Reason (from the status table) |
|---|---|---|---|
| 13BD | E3-CryoFold | failed | prealign: exit code 1; the model file 13BD_e3_prealign.pdb was not written |
| 26DK | MICA | failed | 3_build: exit code -9; the output of 3_build was not written |
| 26DK | EModelX(+AF) | failed | temp_flex: killed by the system (exit code -9, probably out of memory); the model file 26dk_EModelX(+AF)_all_atom_model.pdb was not written |
| 26DK | E3-CryoFold | failed | prealign: killed by the system (exit code -9, probably out of memory); the model file 26DK_e3_prealign.pdb was not written |
| 9OC0 | MICA | failed | 3_build: timeout after 7200 s; the output of 3_build was not written |
| 9OC0 | E3-CryoFold | failed | prealign: exit code 1; the model file 9OC0_e3_prealign.pdb was not written |
| 9OTA | E3-CryoFold | failed | prealign: exit code 1; the model file 9OTA_e3_prealign.pdb was not written |
| 9VI4 | MICA | failed | 3_build: exit code -9; the output of 3_build was not written |
| 9VI4 | ModelAngelo | failed | build: timeout: the run limit of 10800 s; the model file 9VI4_modelangelo.cif was not written |
| 9VI4 | EModelX(+AF) | failed | temp_flex: killed by the system (exit code -9, probably out of memory); the model file 9vi4_EModelX(+AF)_all_atom_model.pdb was not written |
| 9VI4 | EMProt | failed | build: killed by the system (exit code -9, probably out of memory); the model file output.cif was not written |
| 9VI4 | E3-CryoFold | failed | prealign: killed by the system (exit code -9, probably out of memory); the model file 9VI4_e3_prealign.pdb was not written |

**Table A9.** Failed runs scored as zero: mean TM-score over all 20 entries and the paired comparison with MICA. The two policies give the same means and tests because every run was either scored or failed; the intervals can differ in the last digit because the bootstrap draws differ.

| Method | $n$ | Mean TM-score [95% CI] | Mean diff. to MICA | Wilcoxon $p$ | Holm $p$ |
|---|---|---|---|---|---|
| *Failed and unusable runs scored 0 (runs not made are left out)* | | | | | |
| MICA | 20 | 0.80 [0.64, 0.93] | – | – | – |
| ModelAngelo | 20 | 0.76 [0.62, 0.88] | −0.04 | 0.738 | 1.000 |
| EModelX(+AF) | 20 | 0.84 [0.70, 0.94] | 0.05 | 0.551 | 1.000 |
| EMProt | 20 | 0.91 [0.80, 0.98] | 0.12 | <0.001 | <0.001 |
| E3-CryoFold | 20 | 0.65 [0.46, 0.81] | −0.15 | 0.002 | 0.005 |
| *Every missing result scored 0* | | | | | |
| MICA | 20 | 0.80 [0.64, 0.93] | – | – | – |
| ModelAngelo | 20 | 0.76 [0.62, 0.88] | −0.04 | 0.738 | 1.000 |
| EModelX(+AF) | 20 | 0.84 [0.70, 0.94] | 0.05 | 0.551 | 1.000 |
| EMProt | 20 | 0.91 [0.80, 0.98] | 0.12 | <0.001 | <0.001 |
| E3-CryoFold | 20 | 0.65 [0.47, 0.81] | −0.15 | 0.002 | 0.005 |

**Table A10.** MICA with and without docked AF3 input on the entries that ran in both set-ups (difference = docked minus no docked input).

| Entry | Res. (Å) | Docking | TM docked | TM no docked input | Difference |
|---|---|---|---|---|---|
| 9JKK | 3.00 | full | 0.995 | 0.991 | +0.004 |
| 9WFL | 2.75 | full | 0.954 | 0.954 | 0.000 |
| 9ZLH | 2.14 | full | 0.994 | 0.994 | 0.000 |
| 9VUH | 2.70 | full | 0.956 | 0.909 | +0.047 |
| 9OTA | 3.20 | full | 0.913 | 0.913 | +0.001 |
| 9KQM | 3.50 | full | 0.929 | 0.856 | +0.073 |
| 9WME | 3.20 | full | 0.941 | 0.929 | +0.012 |
| 9RWS | 3.28 | full | 0.943 | 0.942 | +0.001 |
| Mean difference 0.017 [0.002, 0.037]; median 0.002; Wilcoxon $p = 0.039$; higher / lower / equal: 6 / 2 / 0 | | | | | |

**Table A11.** TM-score of every method on every entry (normalised by the deposited model; F: failed run).

| Entry | Res. (Å) | MICA | ModelAngelo | EModelX(+AF) | EMProt | E3-CryoFold |
|---|---|---|---|---|---|---|
| *Resolution ≤ 3.0 Å* | | | | | | |
| 9T5K | 2.60 | 0.80 | 1.00 | 0.99 | 1.00 | 0.68 |
| 22FX | 1.24 | 0.98 | 1.00 | 0.97 | 1.00 | 0.97 |
| 9JKK | 3.00 | 0.99 | 0.99 | 0.99 | 1.00 | 0.98 |
| 9TKA | 2.70 | 0.99 | 1.00 | 0.99 | 1.00 | 0.98 |
| 9WFL | 2.75 | 0.95 | 0.99 | 0.95 | 0.99 | 0.98 |
| 9VI4 | 1.80 | F | F | F | F | F |
| 9ZLH | 2.14 | 0.99 | 1.00 | 0.99 | 0.99 | 0.96 |
| 9VUH | 2.70 | 0.96 | 0.75 | 0.91 | 0.97 | 0.63 |
| 9OC0 | 2.97 | F | 0.78 | 0.86 | 0.67 | F |
| 13BD | 2.65 | 0.88 | 0.98 | 0.92 | 0.99 | F |
| *Resolution > 3.0 to 4.0 Å* | | | | | | |
| 9WQ6 | 3.40 | 0.90 | 0.45 | 0.89 | 0.94 | 0.80 |
| 9V4D | 3.07 | 0.94 | 0.58 | 0.94 | 0.95 | 0.91 |
| 9OTA | 3.20 | 0.91 | 0.91 | 0.93 | 0.98 | F |
| 9KQM | 3.50 | 0.93 | 0.79 | 0.86 | 0.96 | 0.90 |
| 9WME | 3.20 | 0.94 | 0.56 | 0.84 | 0.95 | 0.60 |
| 9VBD | 3.26 | 0.98 | 0.24 | 0.99 | 0.99 | 0.97 |
| 9RWS | 3.28 | 0.94 | 0.97 | 0.96 | 0.98 | 0.96 |
| 9WTX | 3.37 | 0.93 | 0.27 | 0.93 | 0.98 | 0.73 |
| 9Z24 | 3.10 | 0.90 | 0.90 | 0.91 | 0.91 | 0.89 |
| 26DK | 3.50 | F | 0.95 | F | 0.98 | F |

**Table A12.** Median (interquartile range) of every measure (Table 2) over the scored models of each method. The sets of entries differ between methods.

| Metric | MICA | ModelAngelo | EModelX(+AF) | EMProt | E3-CryoFold |
|---|---|---|---|---|---|
| TM-score | 0.94 (0.91–0.98) | 0.91 (0.67–0.99) | 0.94 (0.91–0.98) | 0.98 (0.96–0.99) | 0.91 (0.77–0.97) |
| C$\alpha$ match (%) | 94.9 (86.0–96.3) | 91.1 (48.0–98.5) | 94.1 (90.1–97.6) | 57.4 (21.8–90.2) | 80.9 (69.2–96.9) |
| Sequence match (%) | 98.5 (96.5–99.7) | 98.8 (98.1–99.8) | 97.4 (87.3–99.4) | 98.5 (96.5–99.7) | 96.0 (75.5–98.3) |
| Coverage | 1.02 (1.00–1.05) | 1.01 (0.89–2.06) | 1.01 (0.98–1.04) | 1.45 (1.08–4.59) | 1.03 (1.00–1.16) |
| C$\alpha$ quality (%) | 96.0 (94.2–97.9) | 96.7 (80.5–99.5) | 94.7 (91.4–98.1) | 98.2 (94.6–99.7) | 92.0 (80.0–98.9) |
| Aligned sequence identity (%) | 99.3 (97.8–99.7) | 99.5 (98.3–99.9) | 98.7 (96.3–99.6) | 99.5 (97.5–100.0) | 98.3 (86.2–99.8) |
| RMSD (Å) | 1.10 (0.68–1.28) | 0.63 (0.39–0.94) | 1.02 (0.78–1.66) | 0.92 (0.41–1.30) | 1.44 (1.05–2.40) |
| Aligned length | 409 (265–495) | 306 (179–468) | 428 (276–501) | 393 (260–513) | 309 (252–470) |

**Table A13.** TM-score by resolution stratum: entries, scored models, median (interquartile range) and mean with 95 % bootstrap CI.

| Method | Entries | Models | Median (IQR) | Mean [95% CI] |
|---|---|---|---|---|
| *Resolution ≤ 3.0 Å* | | | | |
| MICA | 10 | 8 | 0.97 (0.93–0.99) | 0.94 [0.89, 0.98] |
| ModelAngelo | 10 | 9 | 0.99 (0.98–1.00) | 0.94 [0.87, 1.00] |
| EModelX(+AF) | 10 | 9 | 0.97 (0.92–0.99) | 0.95 [0.92, 0.98] |
| EMProt | 10 | 9 | 0.99 (0.99–1.00) | 0.96 [0.88, 1.00] |
| E3-CryoFold | 10 | 7 | 0.97 (0.82–0.98) | 0.88 [0.77, 0.98] |
| *Resolution > 3.0 to 4.0 Å* | | | | |
| MICA | 10 | 9 | 0.93 (0.91–0.94) | 0.93 [0.92, 0.95] |
| ModelAngelo | 10 | 10 | 0.68 (0.47–0.91) | 0.66 [0.49, 0.82] |
| EModelX(+AF) | 10 | 9 | 0.93 (0.89–0.94) | 0.92 [0.89, 0.94] |
| EMProt | 10 | 10 | 0.97 (0.95–0.98) | 0.96 [0.95, 0.98] |
| E3-CryoFold | 10 | 8 | 0.90 (0.78–0.92) | 0.85 [0.76, 0.92] |

**Table A14.** Fragment-agnostic TM-score (US-align `-mm 5`): median (interquartile range) over the scored models, and the paired comparison with MICA on the entries where both built a model (method minus MICA; Wilcoxon $p$ and its Holm correction over the four methods).

| Method | $n$ | TM-score, fragment-agnostic, median (IQR) | Mean difference to MICA [95% CI] | Wilcoxon $p$ | Holm $p$ |
|---|---|---|---|---|---|
| MICA | 17 | 0.96 (0.94–0.98) | – | – | – |
| ModelAngelo | 19 | 0.96 (0.82–0.99) | −0.07 [−0.14, −0.00] | 0.243 | 0.487 |
| EModelX(+AF) | 18 | 0.95 (0.93–0.98) | 0.00 [−0.01, 0.03] | 0.464 | 0.487 |
| EMProt | 19 | 0.99 (0.98–0.99) | 0.03 [0.01, 0.05] | 0.002 | 0.007 |
| E3-CryoFold | 15 | 0.93 (0.84–0.97) | −0.05 [−0.08, −0.02] | 0.002 | 0.007 |

**Table A15.** Further output files of each method, scored with the same protocol next to the file of the main comparison (marked *scored*); $n$ is the number of entries with the file. Only the file marked *scored* is used in the comparisons.

| Output | $n$ | TM-score, median | TM-score, mean | Fragment-agnostic TM, median | Cα match, median | Coverage, median | Cα quality, median |
|---|---|---|---|---|---|---|---|
| MICA, all-atom model (scored) | 17 | 0.94 | 0.94 | 0.96 | 94.9 | 1.02 | 96.0 |
| MICA, Cα model after chain building | 17 | 0.94 | 0.94 | 0.96 | 94.9 | 1.02 | 96.2 |
| MICA, initial Cα model of the network | 17 | 0.82 | 0.82 | 0.86 | 99.5 | 0.86 | 84.4 |
| ModelAngelo, sequence-assigned model (scored) | 19 | 0.91 | 0.79 | 0.96 | 91.1 | 1.01 | 96.7 |
| ModelAngelo, model without sequence assignment | 19 | 0.50 | 0.50 | 0.99 | 48.5 | 1.47 | 98.2 |
| EModelX(+AF), all-atom model (scored) | 18 | 0.94 | 0.94 | 0.95 | 94.1 | 1.01 | 94.7 |
| EModelX(+AF), Cα model | 18 | 0.94 | 0.94 | 0.95 | 94.1 | 1.01 | 94.7 |
| EMProt, build with the AF3 models (scored) | 19 | 0.98 | 0.96 | 0.99 | 57.4 | 1.45 | 98.2 |
| EMProt, de novo model | 19 | 0.96 | 0.82 | 0.97 | 97.2 | 1.00 | 97.1 |
| E3-CryoFold, `pre_align` model (scored) | 15 | 0.91 | 0.86 | 0.93 | 80.9 | 1.03 | 92.0 |

**Table A16.** Spearman rank correlation (two-sided $p$ in brackets, not corrected for multiple tests) of the TM-score with the resolution of the map, the number of modelled residues and the highest identity to older PDB entries; $n$ is the number of scored models.

| Method | $n$ | Resolution | Modelled residues | Identity to older PDB |
|---|---|---|---|---|
| MICA | 17 | −0.25 (0.325) | 0.03 (0.896) | 0.02 (0.940) |
| ModelAngelo | 19 | −0.66 (0.002) | −0.29 (0.237) | 0.05 (0.835) |
| EModelX(+AF) | 18 | −0.48 (0.044) | −0.18 (0.463) | −0.42 (0.080) |
| EMProt | 19 | −0.53 (0.018) | −0.13 (0.609) | −0.28 (0.247) |
| E3-CryoFold | 15 | −0.28 (0.305) | −0.00 (0.990) | −0.11 (0.685) |

**Table A17.** Wall-clock time per run (failed runs included). For MICA it is the time of programs 1 to 3 including docking; the baselines ran with another program on the second GPU of the session, so the times are indicative.

| Method | Runs | Median (min) | Range (min) | Total (h) | Device |
|---|---|---|---|---|---|
| MICA | 20 | 81.6 | 32.8–557.5 | 40.0 | cuda |
| ModelAngelo | 20 | 20.3 | 7.9–180.0 | 12.2 | 0, 1 |
| EModelX(+AF) | 20 | 6.9 | 2.4–96.8 | 4.8 | 0, 1 |
| EMProt | 20 | 14.2 | 4.6–58.5 | 7.0 | 0, 1 |
| E3-CryoFold | 20 | 2.9 | 1.0–20.3 | 1.7 | 0, 1 |

**Table A18.** Paired comparison of the TM-score with MICA run without docked input, on the entries where that run and the method both have a model (method minus MICA; as Table 5).

| Method | Pairs | Mean difference [95% CI] | Median diff. | Wilcoxon $p$ | Holm $p$ | $t$-test Holm $p$ | higher / lower / equal |
|---|---|---|---|---|---|---|---|
| *All entries* | | | | | | | |
| ModelAngelo | 8 | −0.07 [−0.17, 0.01] | 0.00 | 0.461 | 1.000 | 0.663 | 4 / 4 / 0 |
| EModelX(+AF) | 8 | −0.01 [−0.03, 0.01] | 0.00 | 0.617 | 1.000 | 0.663 | 4 / 4 / 0 |
| EMProt | 8 | 0.04 [0.02, 0.07] | 0.04 | 0.016 | 0.062 | 0.041 | 7 / 1 / 0 |
| E3-CryoFold | 7 | −0.08 [−0.20, 0.02] | −0.01 | 0.578 | 1.000 | 0.663 | 3 / 4 / 0 |
| *Resolution ≤ 3.0 Å* | | | | | | | |
| ModelAngelo | 4 | −0.03 [−0.12, 0.03] | 0.00 | 0.875 | 1.000 | 1.000 | 3 / 1 / 0 |
| EModelX(+AF) | 4 | 0.00 [−0.00, 0.00] | 0.00 | 0.750 | 1.000 | 1.000 | 1 / 3 / 0 |
| EMProt | 4 | 0.03 [0.00, 0.05] | 0.02 | 0.250 | 1.000 | 0.738 | 3 / 1 / 0 |
| E3-CryoFold | 4 | −0.07 [−0.21, 0.01] | −0.02 | 0.375 | 1.000 | 1.000 | 1 / 3 / 0 |
| *Resolution > 3.0 to 4.0 Å* | | | | | | | |
| ModelAngelo | 4 | −0.10 [−0.28, 0.01] | −0.04 | 0.375 | 1.000 | 1.000 | 1 / 3 / 0 |
| EModelX(+AF) | 4 | −0.01 [−0.06, 0.02] | 0.01 | 0.875 | 1.000 | 1.000 | 3 / 1 / 0 |
| EMProt | 4 | 0.06 [0.03, 0.09] | 0.05 | 0.125 | 0.500 | 0.165 | 4 / 0 / 0 |
| E3-CryoFold | 3 | −0.09 [−0.33, 0.04] | 0.02 | 1.000 | 1.000 | 1.000 | 2 / 1 / 0 |

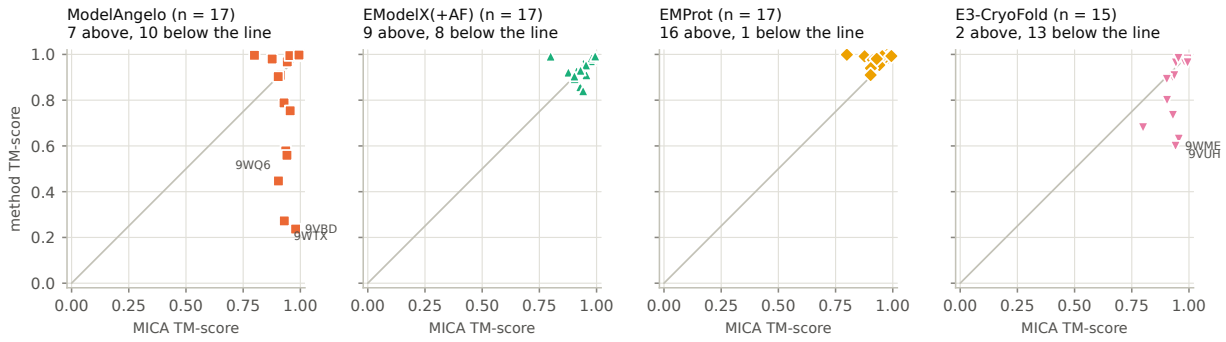



**Figure A2.** TM-score of each method against that of MICA on the same entry (the line is equality; entries that differ by 0.3 or more are labelled, at most three per panel).

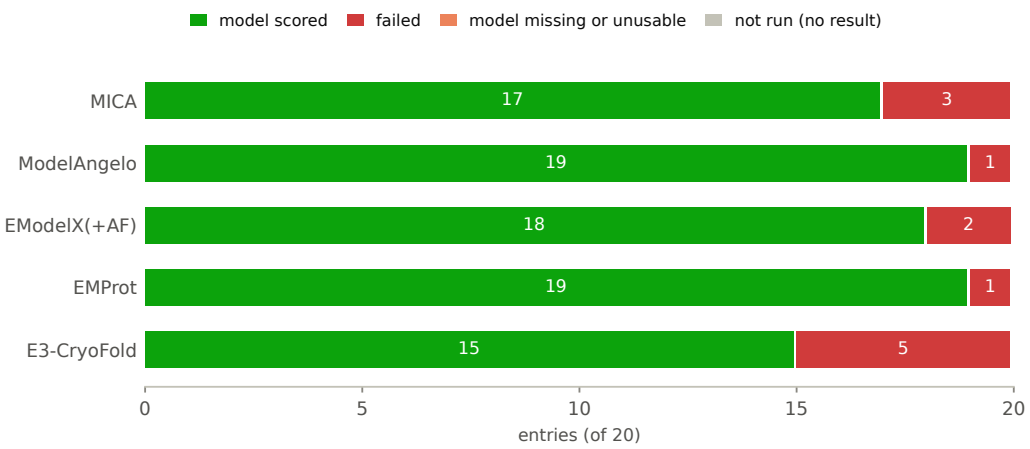



**Figure A3.** Outcome of the 20 runs of each method.

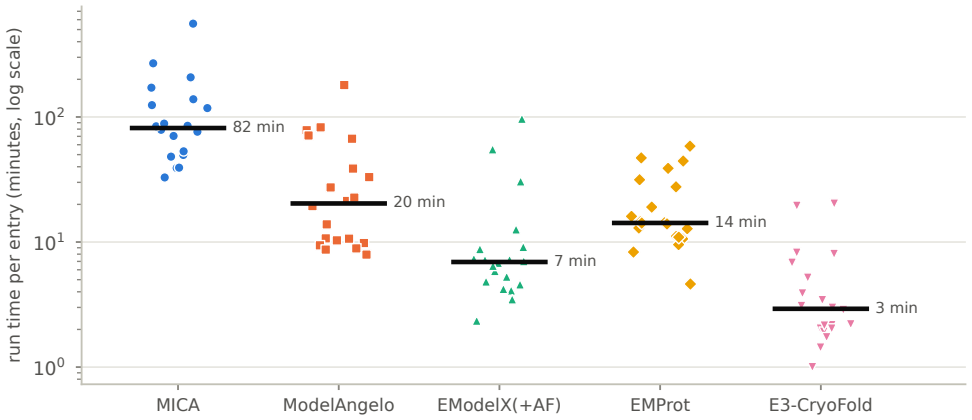



**Figure A4.** Run time of every run (failed runs included; logarithmic axis); the bar is the median.

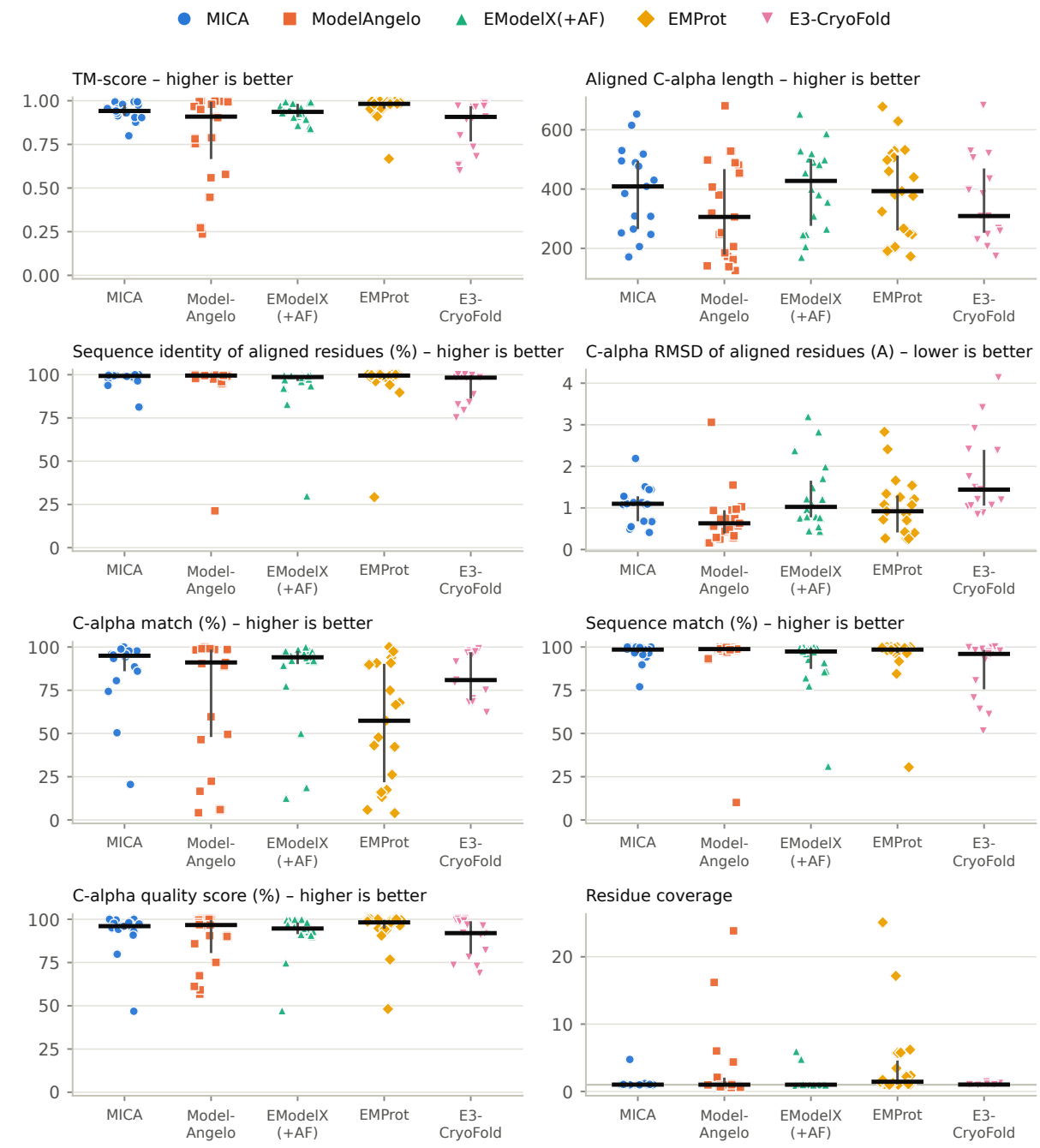



**Figure A5.** All measures over the scored models of each method (as Figure A1).

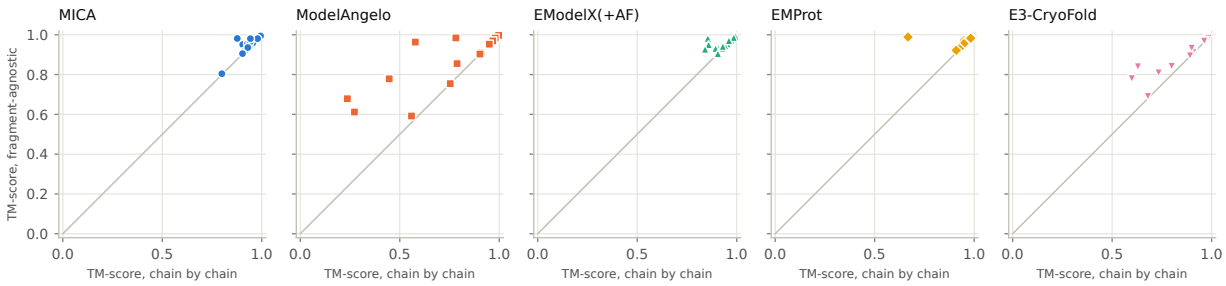



**Figure A6.** TM-score chain by chain (the primary measure) against the fragment-agnostic TM-score. A point above the line is a model whose primary score is lowered by fragmentation or by extra chains.

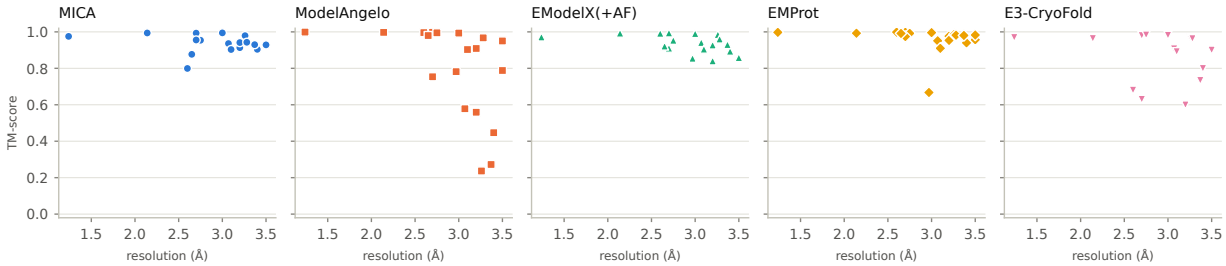



**Figure A7.** TM-score of each method against the resolution of the map.

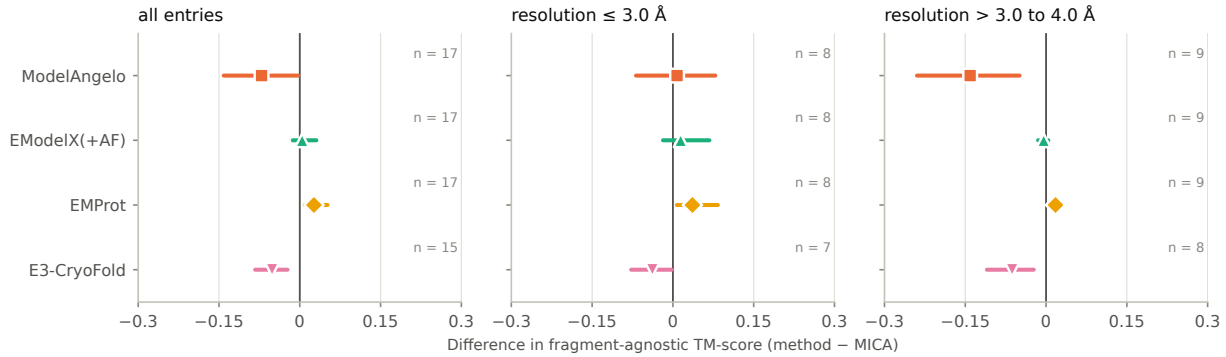



**Figure A8.** Difference between each method and MICA in the fragment-agnostic TM-score (as Figure 2).